

PM6225/PM6375 Power Management IC

80-27912-5 Rev. AC

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Revision History

Revision	Date	Description
AA	June 2021	Initial release
AB	September 2021	■ Slide 10 <i>PM6225/PM6375 Functional Block Diagram (design example)</i>: Updated this slide ■ Slide 17 <i>FTS510 SMPS – Schematic Example</i>: Updated this slide ■ Slide 24 <i>HFS510 SMPS – Schematic Example</i>: Updated this slide ■ Slide 29 <i>Subregulation for SM6225</i>: Updated this slide ■ Slide 50 <i>UVLO</i>: Updated this slide ■ Slide 51 <i>Overvoltage Lockout (OVLO)</i>: Updated this slide ■ Slide 58 <i>Clocks</i>: Added this section ■ Slide 68 <i>PM6375 and PM6225 Layout Guidelines</i>: Added this section
AC	July 2023	Global: Updated the document for SXR1010P

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Section 1

Power Management System and IC Overview

1.1 Reference Documents

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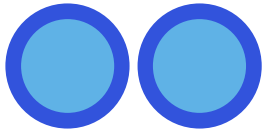
1.2 IC Overview

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Power Management System

- PM6225 is optimized for the SM6225 chipset
- PM6375 is optimized for the SM4350/SM6375 and SXR1010P chipsets
- System-clock and sleep-clock sources for entire chipset
 - Two RF (low-noise) outputs
 - Three baseband (low-noise) output
- Sleep clock output

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Section 1.1

Reference Documents

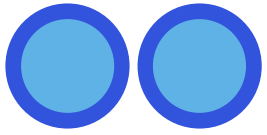
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PMIC Reference Documents

DCN	Document title
Power management documents	
80-27912-1	PM6225/PM6375 Power Management IC Data Sheet
80-27912-4	PM6225/PM6375 Device Revision Guide
80-27912-5	PM6225/PM6375 Power Management IC Design Guidelines/Training Slides (this document)

NOTE: Released documents are posted at <https://createpoint.qti.qualcomm.com> (the CreatePoint website) and are available for download.

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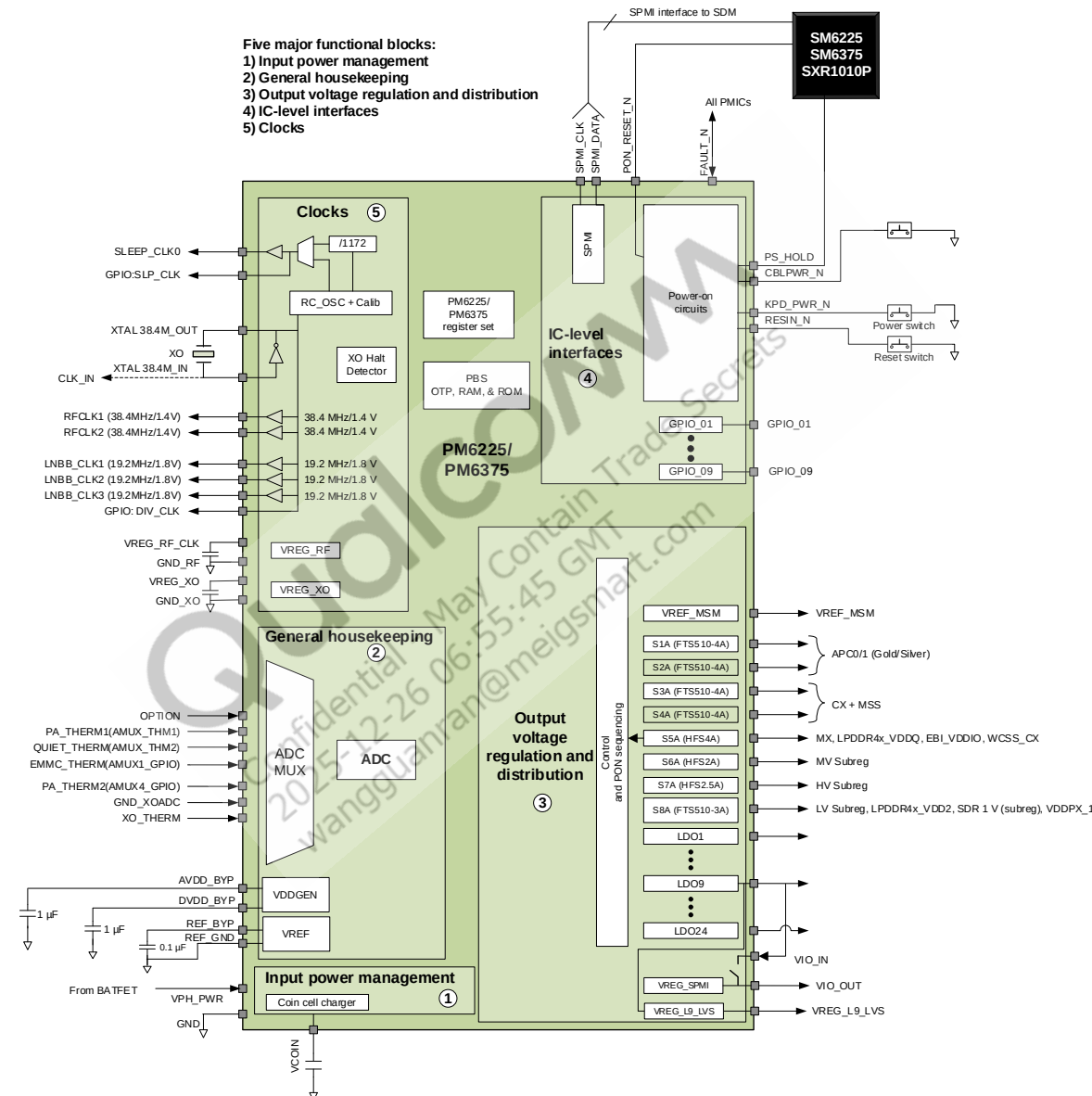


Section 1.2

IC Overview

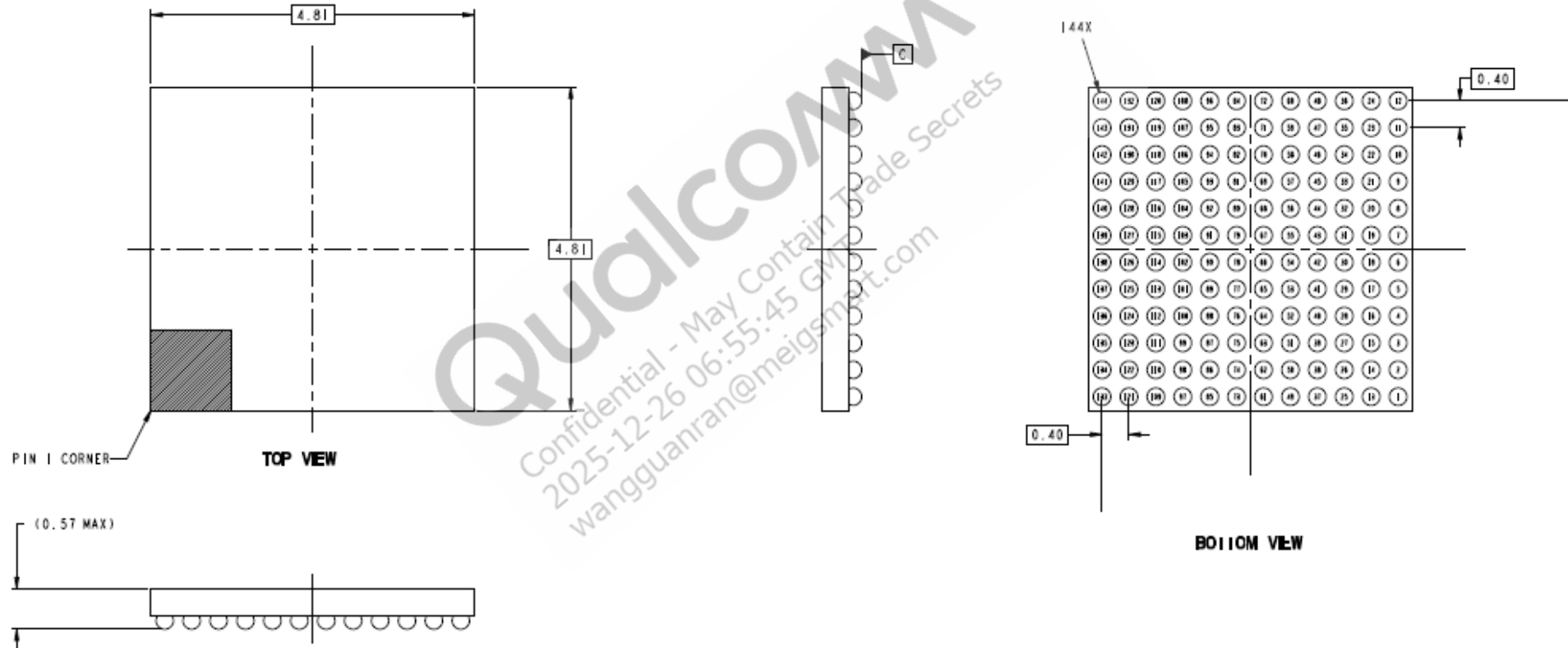
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PM6225/PM6375 Functional Block Diagram (Design Example)



PM6225/PM6375 Package – FOWNSP144

- 4.81 mm × 4.81 mm
- 0.57 mm maximum height



PM6225/PM6375 Features

Feature	PM6225/PM6375 capability
Input power management	
Coin cell or capacitor backup	Keep-alive power source; orchestrated charging
Output voltage regulation	
Switched-mode power supplies HF-SMPS FT-SMPS	Three Five
Low-dropout linear regulators	24
Pseudo-capless LDO designs	15
General housekeeping	
On-chip ADC	Shared housekeeping (HK) and XO support
Analog multiplexing for ADC HK inputs XO input	Many internal nodes and external inputs Dedicated pin (XO_THERM)
Overtemperature protection	Multistage smart thermal control
38.4 MHz oscillator support	XO (with on-chip ADC)
XO controller and XO outputs	Five sets: three low-noise baseband outputs and two low-power RF outputs
Special purpose clock outputs	Sleep clock; 19.2 MHz, 9.6 MHz, 4.8 MHz, 2.4 MHz, and 1.2 MHz, including low-power mode 2.4 MHz for MP3; three high-speed GPIOs for fast clocks
Real-time clock (RTC)	RTC clock circuits and alarms
IC-level interfaces	
Primary status and control	Two-line SPMI
Interrupt managers	Supported by SPMI
Power sequencing	Power-on, power-off, and resets

PM6225/PM6375 Features (cont.)

Feature	PM6225/PM6375 capability
Configurable I/Os	
GPIO pins	Nine; configurable as digital inputs or outputs or analog pass-through
Package	
Size	4.81 × 4.81 × 0.57 mm
Pin count and package type	FOWNSP144

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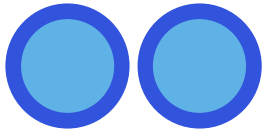
Output Power Management

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PM6225/PM6375 Regulator Summary

See *PM6225/PM6375 Power Management IC Data Sheet* (80-27912-1) for the regulator summary.

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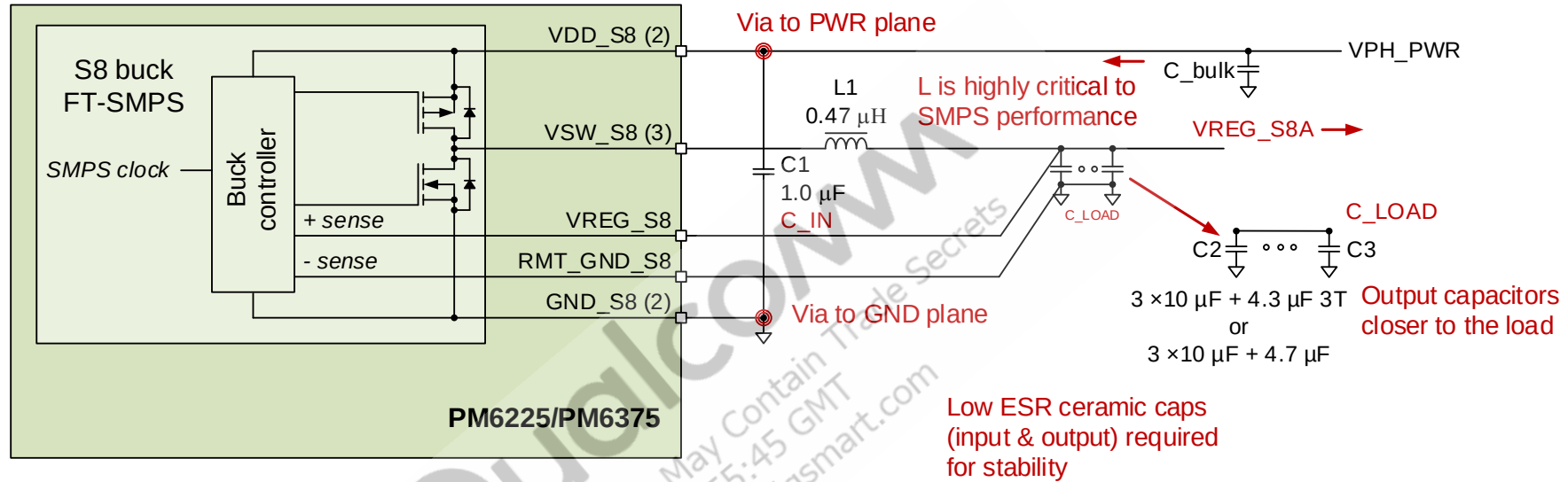


Section 2.1

Fast-Transient SMPS

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FTS510 SMPS – Schematic Example



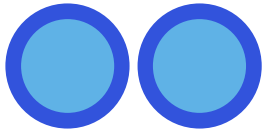
NOTES:

- See the SM6375 + PM6375 + PM7250B + PMK8003 + SMR526 Reference Schematic (80-26086-41) for all schematic recommendations.
- See the SM6225 + PM6225 + PM7250B + LPDDR4X Reference Schematic (80-26896-41) for all schematic recommendations.

Fast-transient SMPS – Operational Details and Features

- Tight regulation, extremely fast transient response.
- Transients that are too fast for linear, duty-cycle control trigger an immediate, nonlinear (100% or 0%) response for fast and inherently smooth transitions back into linear control.
- **Major benefit of FT-SMPS design: ability to support multiphase operation with an arbitrary and configurable number of phases. Multiphase operation increases the maximum current and transient capability.**
- FT-SMPS operates in PWM only when in active mode.

NOTE: See the *Application Note: Understanding High-frequency and Fast-transient Switched-mode Power Supplies* (80-VT310-124) for more details.

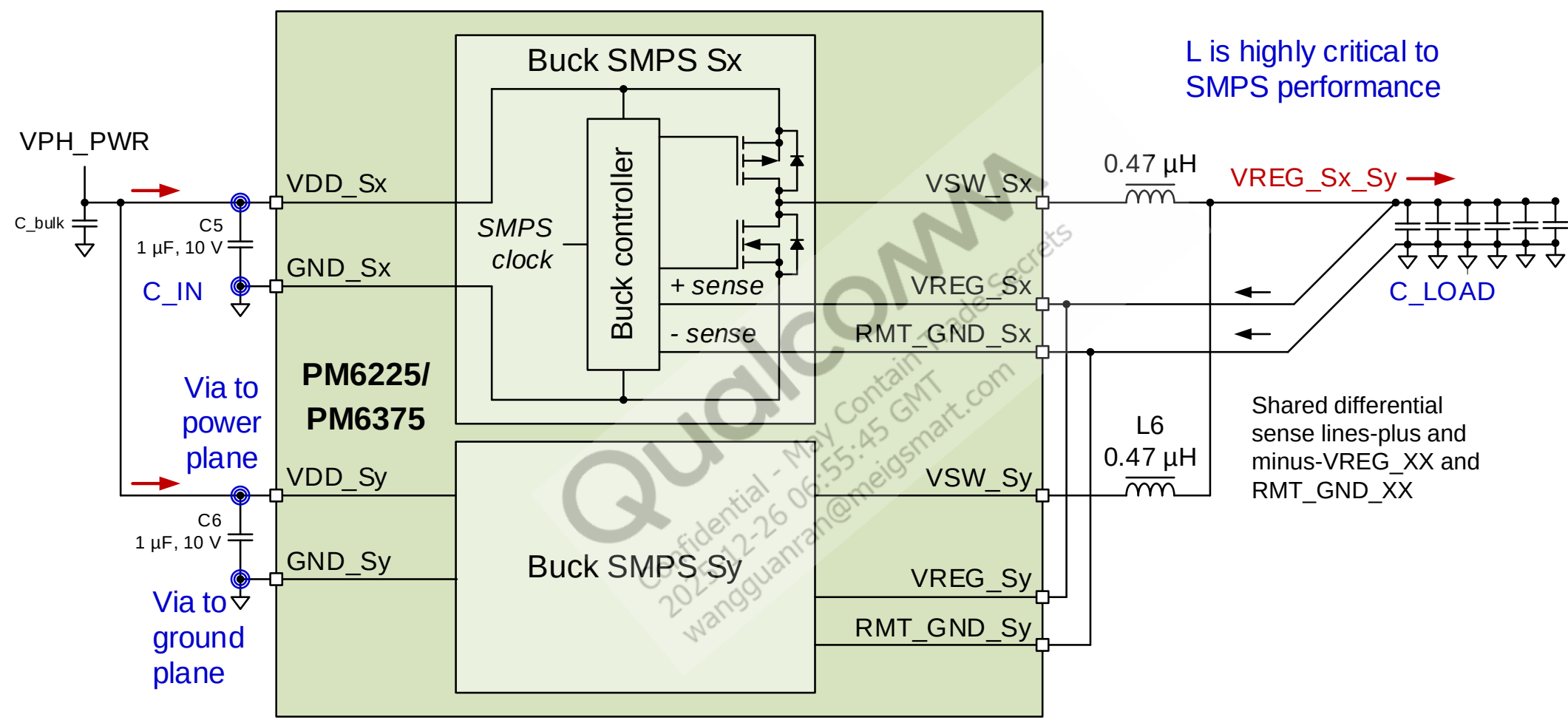


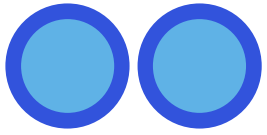
Section 2.2

Multiphase SMPS

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Multiphase SMPS





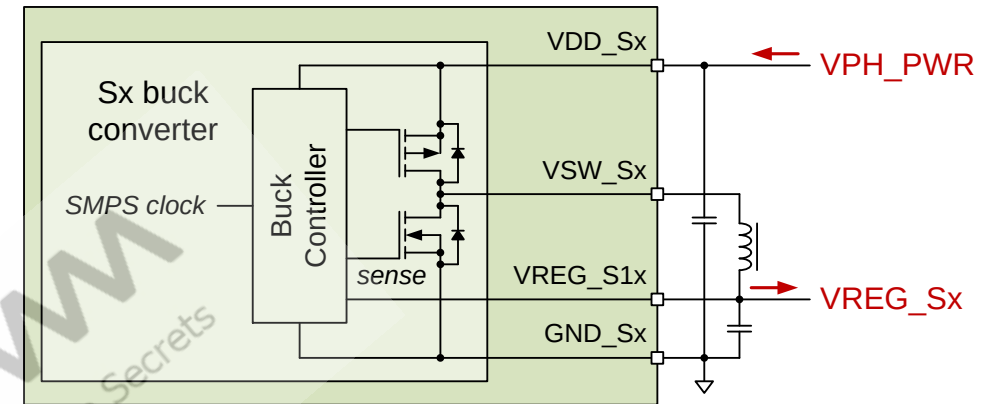
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High-Frequency SMPS

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HF-SMPS Features and Operation

- S5, S6, and S7 are HF-SMPS
- Good transient response
- Slow start feature
- Current limiting
- Overcurrent protection (OCP)
- Operating modes
 - PWM
 - Current-mode constant-frequency PWM control
 - Delivers the specified rated current to the load
 - Pulse skipping
 - PFM
 - The power switch is only turned on when the output voltage dips below a threshold.
 - Main advantage: maintains high efficiency even at light loads.
 - Auto mode
 - Automatic switching between PFM and PWM modes, based on load current
 - Programmable threshold



NOTE: See the *Application Note: Understanding High-frequency and Fast-transient Switched-mode Power Supplies (80-VT310-124)* for more details.

SMPS Buck OCP

The OCP feature has been implemented to avoid inductor current runaway in low impedance faults.

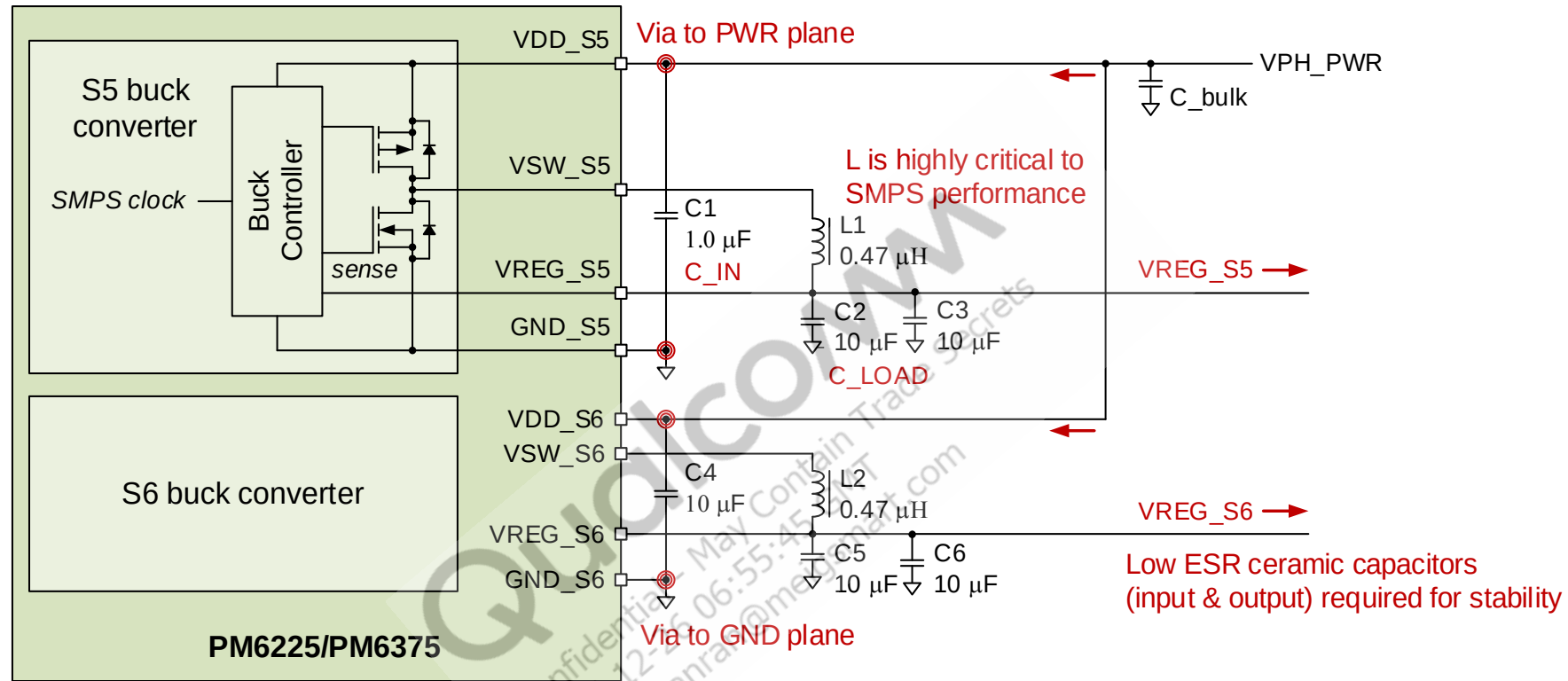
- When output voltage falls ~100 mV below the set point every time SMPS hits the current limit threshold, then following occurs:
 - a. High side FET is turned off and if current does not reach zero before INZERO timer expires, then buck is turned on.
 - b. After the buck turns on and if the buck again reaches current limit, then repeat step A until output voltage is ~100 mV above the set point. Else buck is turned off after OCP timer expires.

The default INZERO timer is 80 μ s and OCP timer is 640 μ s.

- The OCP turns off the buck to protect it against damage.
- To recover, the SMPS must be reset (disable and then enable it).

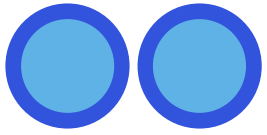
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HFS510 SMPS – Schematic Example



NOTES:

- See the SM6375 + PM6375 + PM7250B + PMK8003 + SMR526 Reference Schematic (80-26086-41) for all schematic recommendations.
- See the SM6225 + PM6225 + PM7250B + LPDDR4X Reference Schematic (80-26896-41) for all schematic recommendations.



Section 2.4

Linear LDO Regulators

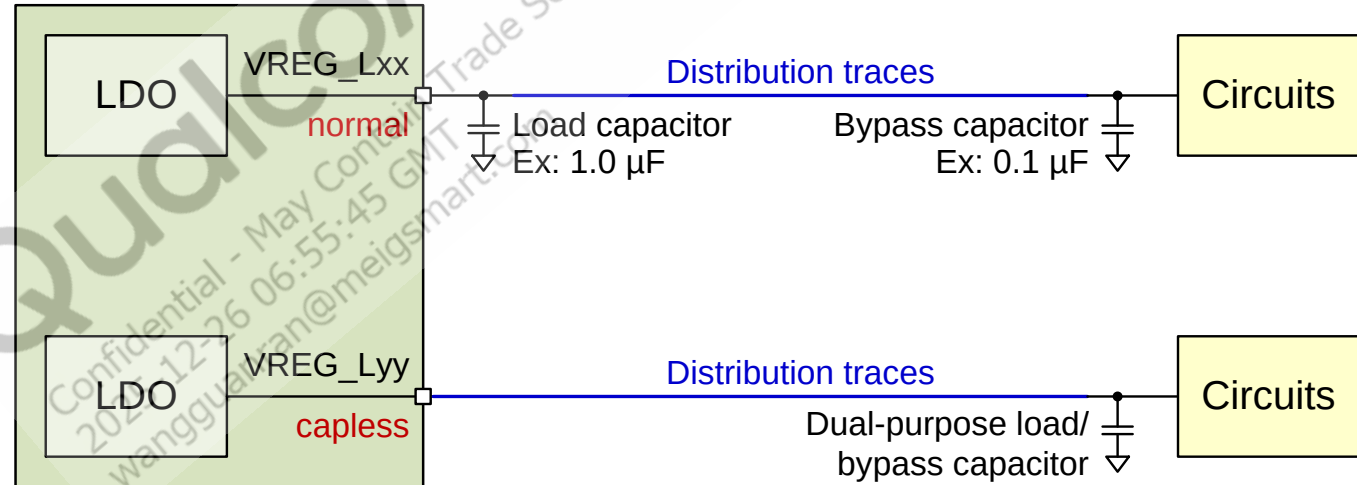
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LDO Types (Nearby Load Capacitors or Pseudocapless) and the LDO LPM

- LDOs that do not require load capacitors near the PMIC are called pseudocapless LDOs.
 - The capacitors at the circuits powered by pseudocapless LDOs are sufficient; additional local capacitors are not required.
 - Pseudocapless LDOs have PCB routing requirements that must be met in order to remove the local capacitor.
- VREG_XO circuits and VREG_RF_CLK clock buffers are used to power the internal PMIC circuits only – they do not support external circuits so their load capacitors must be located near the PMIC.
- LDO L9 powers internal circuit and requires a 10 μF load capacitor near the PMIC.
- LDO L6 powers the DDR memory and requires $2 \times 10 \mu\text{F}$, and $1 \times 10 \mu\text{F}$ DNI load capacitors for supporting the transient response.

- Normal load capacitor situation
- Load capacitor at LDO output
 - Decoupling capacitor at load

- Pseudo-capless situation
- Reduce BOM cost
 - Reduce board area

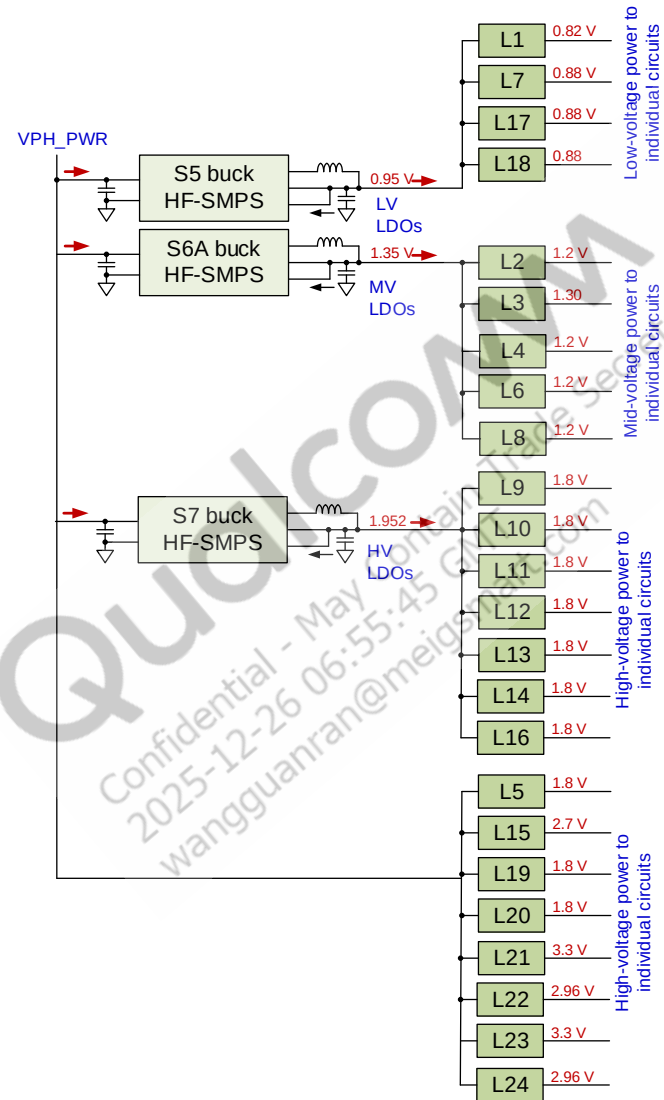


- An LDO implements its LPM by reducing the current of its feedback loop. During low-power operation, the regulator performance is degraded – lower PSRR, less output current capability, and so forth. If the load is greater than 10 mA, the output voltage is likely to be out of specification during LPM.

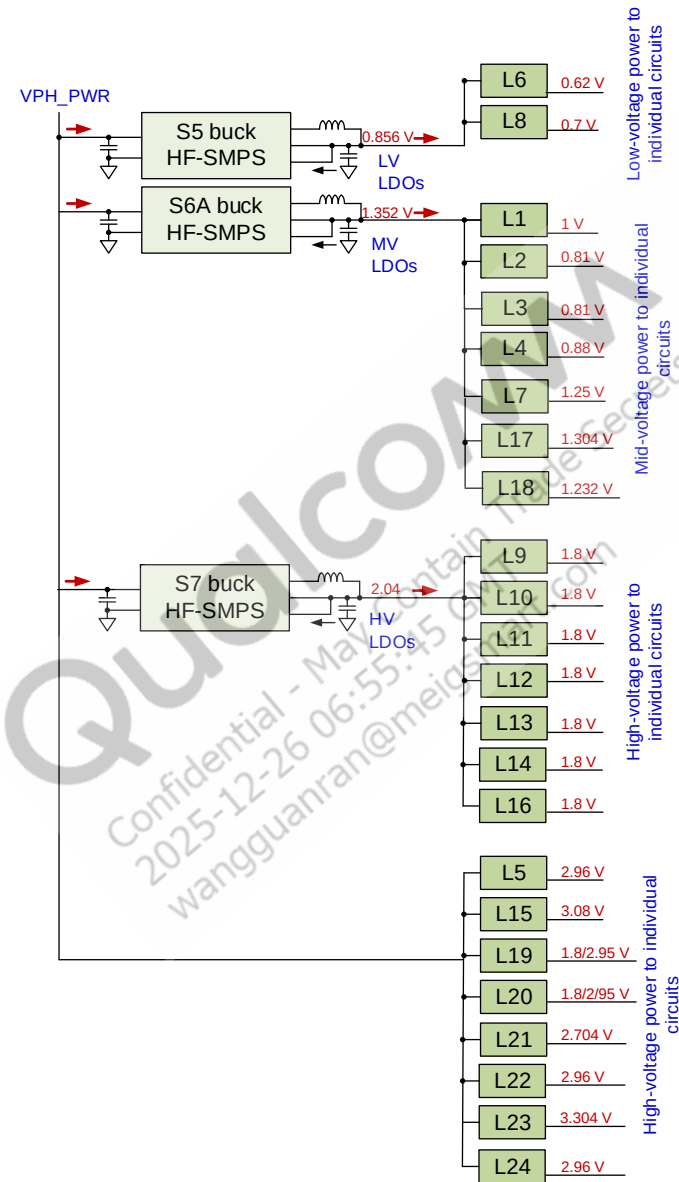
Subregulation

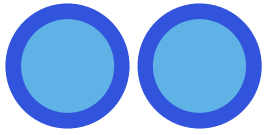
- Subregulation implements a two-stage regulation scheme: the first stage is a buck converter, and the second stage is a linear regulator.
- Rather than dropping a large voltage across an LDO (such as 3.6 V to 1.05 V), the buck converter efficiently creates an intermediate voltage (1.3 V) that is used by several LDOs to create the anticipated outputs.
 - The intermediate voltage (1.3 V, for example) can also be used to power other circuits.
- Low LDO headroom provides greatly reduced power dissipation.
- Subregulation allows fewer SMPS circuits, resulting in reduction of the die area, PWB area, and BOM cost.
- Subregulation provides sufficient suppression of buck noise and spurs to enable the auto mode of the buck.
 - Auto mode yields significantly better buck efficiency when the load is below ~ 100 mA to 150 mA.
- Subregulation also reduces ripple and noise of the supply, resulting in improved circuit performance.
- RFIC supplies use subregulation to improve noise and ripple.
- Dedicated LDOs for different circuits provide isolation between those circuits.
- As shown in the following slides, subregulation is used extensively.

Subregulation for SM6375/SM4350/SXR1010P



Subregulation for SM6225





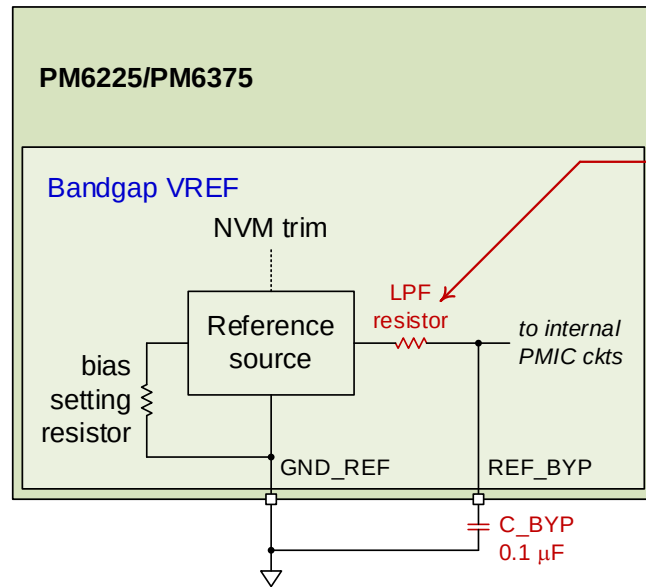
Section 2.5

Master Bandgap (MBG)

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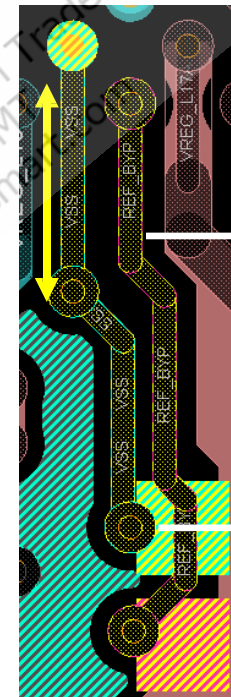
REF_BYP and GND_REF Routing

- REF_BYP capacitor is a main internal reference for the PMIC.
- Route both sides of the capacitor (REF_BYP and REF_GND) to the PMIC balls.
- Connection to the main ground plane should be at the PMIC ball (do not connect the capacitor to the GND plane).



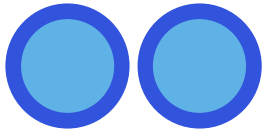
This circuit's ground is brought off-chip at GND_REF; it is isolated from others until they converge on the main ground plane.

Layout example



REF_GND connects to main GND at a very short distance from the PMIC ball (0.83 mm).

REF_BYP and REF_GND are routed on layer 2 between the quiet LDOs on layer 1 and layer 3, with a minor crossing of VREG_S6A powering LDOs.



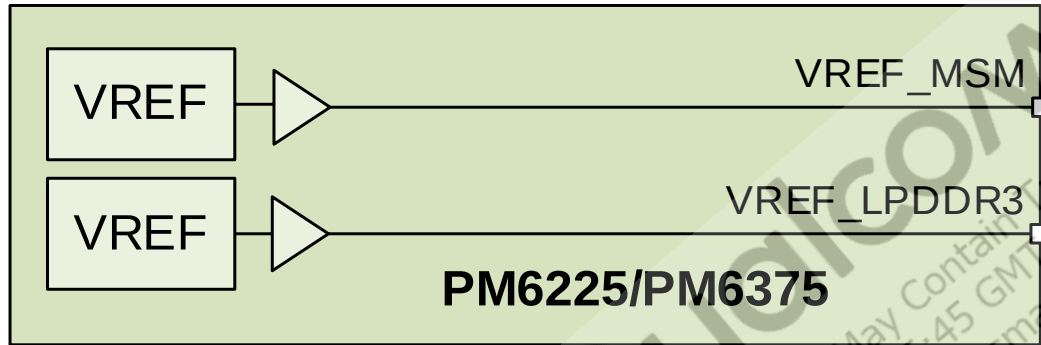
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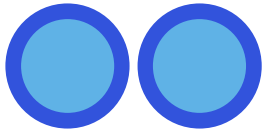
Voltage Reference

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Voltage Reference Outputs

- PM6225/PM6375 has dedicated pins for VREF_LPDDR3 and VREF_MSM chipsets both powered from VPH.
- VREF_LPDDR3 is a 0.6 V reference used only for LPDDR3 and left unconnected otherwise.
- VREF_MSM is 1.25 V reference required for VREF_HVPAD of MSM.





Section 2.7

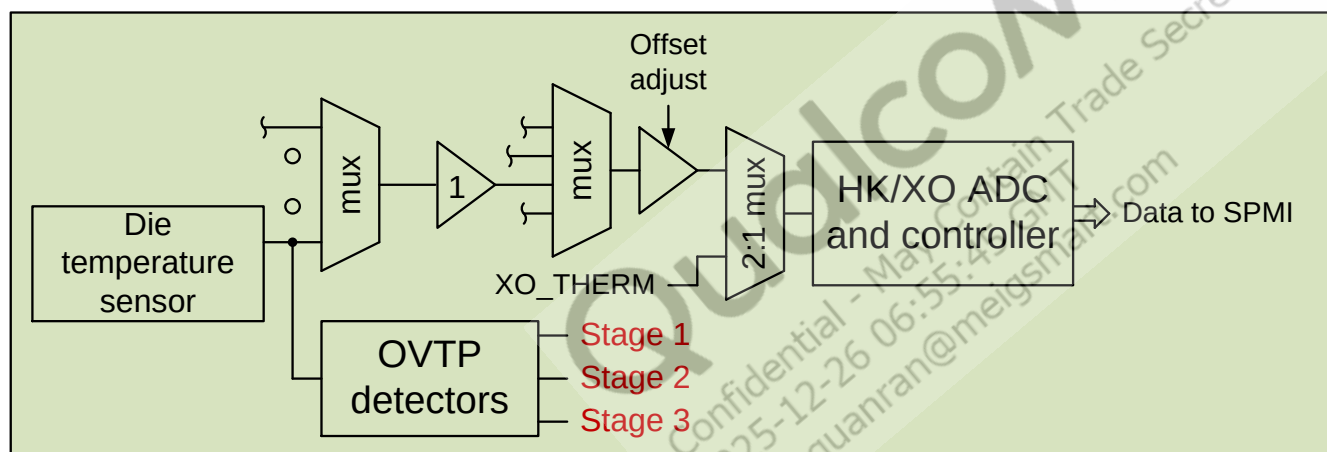
Overtemperature Protection

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Overtemperature Protection

Each PMIC provides overtemperature protection in stages, depending on the level of urgency as the die temperature rises.

- Stage 0: Normal operating conditions (less than 110°C); no interrupt is generated
- Stage 1: 110°C to 130°C; interrupt sent to modem IC without shutting down any PM circuits
- Stage 2: 130°C to 150°C; an interrupt is sent to the modem IC, and high-current drivers (LED drivers, backlight drivers, and so on) are shut down.
- Stage 3: Greater than 150°C; an interrupt is sent to the modem IC, and the PM functions are completely shut down.



Temperature hysteresis is incorporated so the die temperature can cool significantly before the device can be powered on again.

- If any start signals are present while at stage 3, they are ignored until stage 0 is reached
- When the device cools enough to reach stage 0 and a start signal is present, the PM circuits power-up immediately



Section 3

IC-level Interfaces

3.1 Programmable Boot Sequence

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3.2 PON Sequence

[42](#)

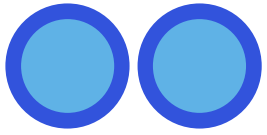
3.3 Resets

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3.4 Coin Cell Support

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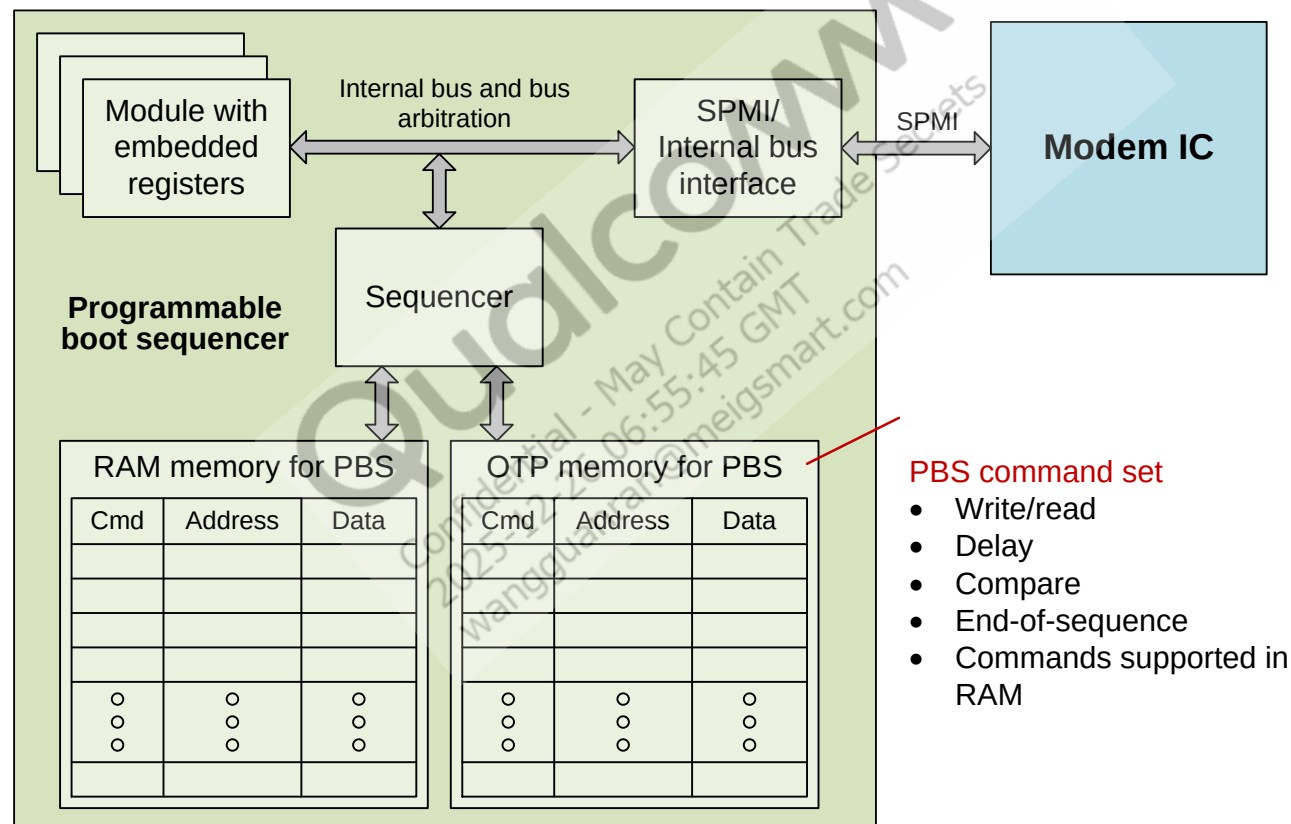
Section 3.1

Programmable Boot Sequence

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PBS

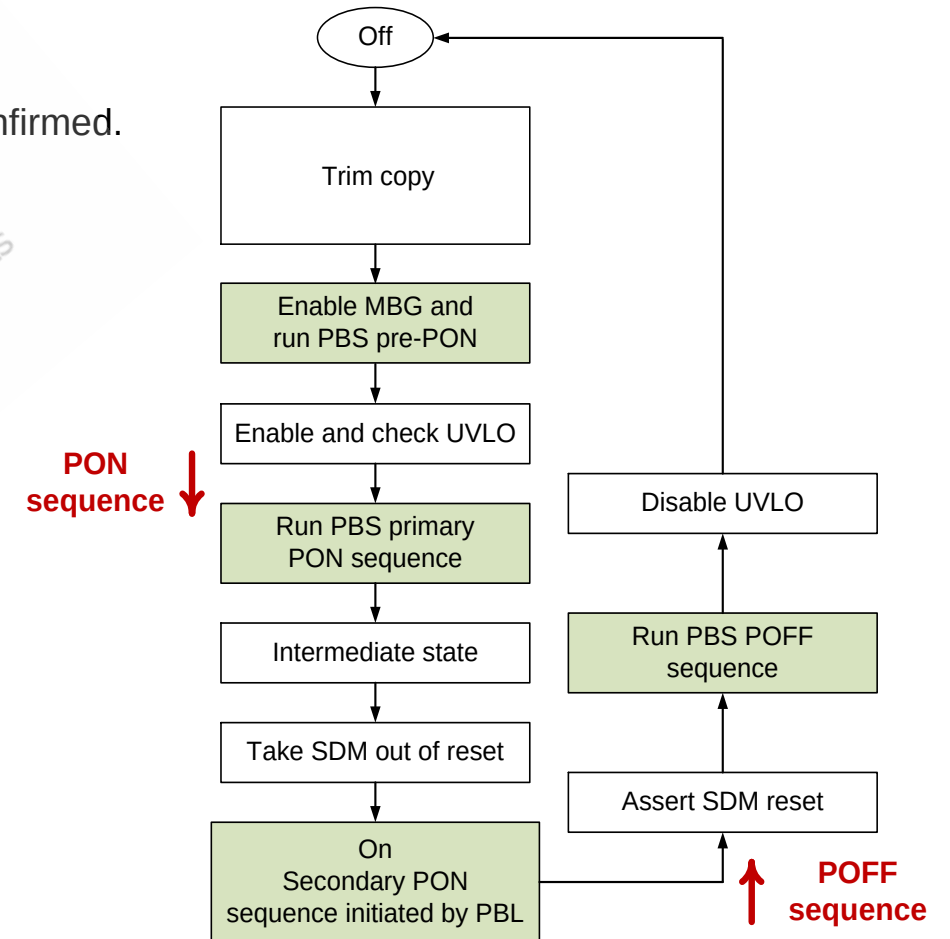
The PM6225/PM6375 devices support boot sequences configured with OTP and RAM. Implemented as a series of address and data transactions that mimic normal software control. Programmable control of any PMIC resource in any order with any available programmable configuration. Support for API-like routines is implemented with OTP-defined sequences; routines are initiated via SPMI writes to address pointer or via hardware trigger. Support for a limited command set of executable functions.



NOTE: Modem IC is applicable for SM6375 and SM6225 only.

Power-on and Power-off

- PBS performs the following during power-on.
 - Configures all the PMIC device features prior to enabling anything in the system.
 - Enables the minimum functions needed to boot the SDM device.
 - Enables processors, eMMC, and so on, once a sufficient battery or charger is confirmed.
- PBS runs once during shutdown for regulator sequencing.



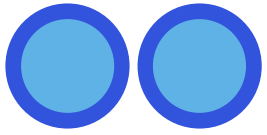
PBS Programming

PBS OTP

- Programmed during ATE by Qualcomm Technologies, Inc. (QTI); customers cannot modify.
- Some areas are reserved for future enhancement or workarounds.

PBS RAM

- For bring-up, programmed during SBL.
- PBS programming must be handled by trusted software.
- Some space is reserved for QTI during development, but these sequences can potentially be moved into OTP prior to commercial sample (CS).



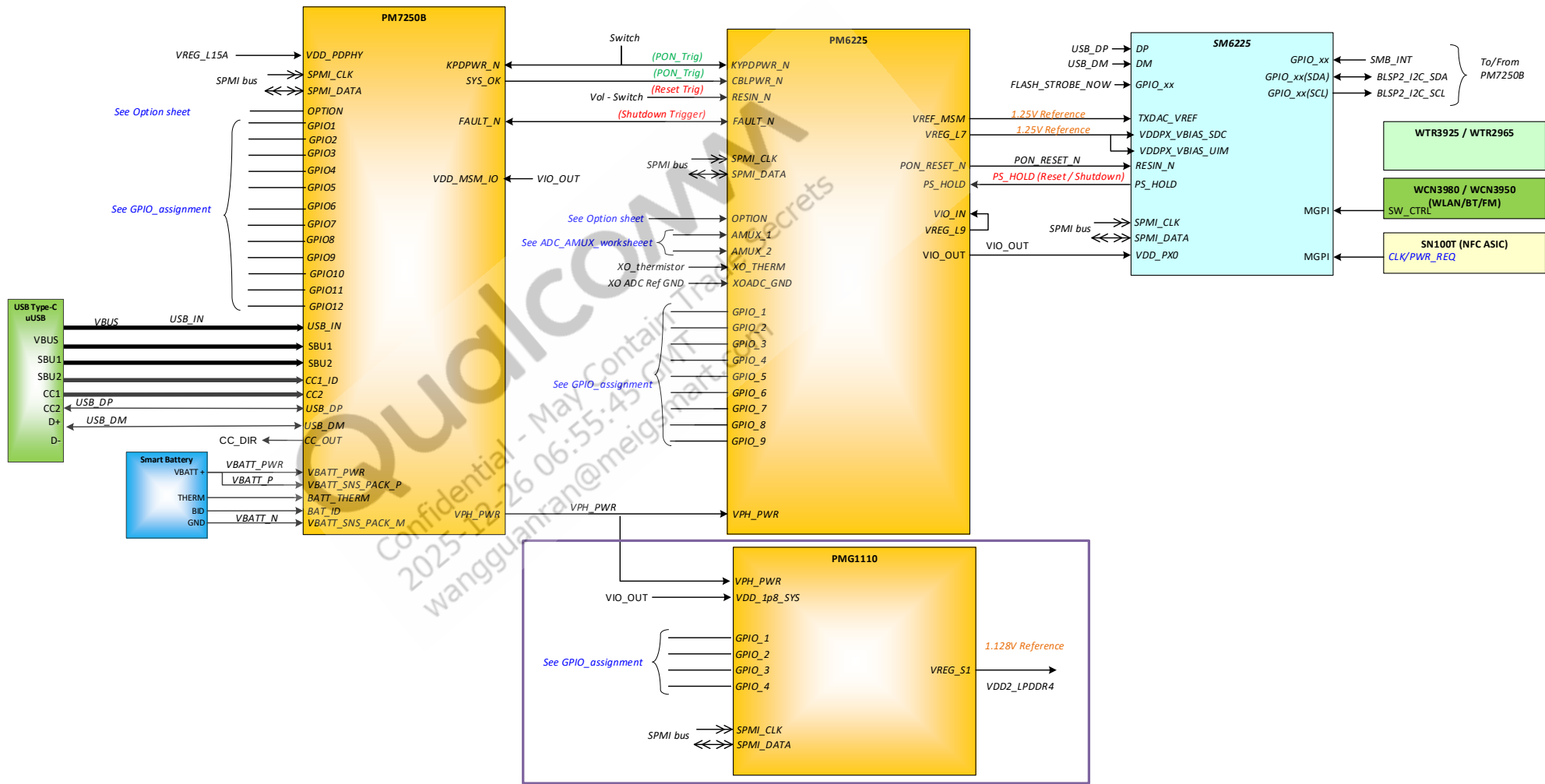
Section 3.2

PON Sequence

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Power-on Circuits

PMIC Interface Diagram for PM6225 + PM7250B + SM6225 Chipsets



Key Power Sequencing Pins – PM6225/PM6375

Pin name	Description	Purpose
KPD_PWR_N	Active-low power-on/off trigger from external button	Initiates the power-on/off sequence of the PM6225/PM6375 device.
PON_RESET_N	Active-low output to the MSM devices to indicate PON complete	Allows the MSM device to come out of reset device to complete its power-on sequence.
VPH_PWR	The PM6225/PM6375 device power	Main power input from the charger PMIC; if voltage is below the UVLO level, power on is not possible.
CBL_PWR_N	Tying this pin to ground initiates a power-on whenever a valid power source is detected (including battery insertion)	To power on when a battery is inserted.
RESIN_N	Active-low power-off trigger from external button	Initiates power reset sequence. A stage 2 and/or stage 3 reset when held at a logic low.

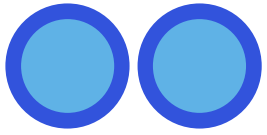
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PM6225/PM6375 Option Pin

- The PM6225/PM6375 device includes one analog optional configuration pin (OPTION) that must be hardwired with resistor connected to the ground.
- A pull-down resistor on the board connected to the OPTION pin sets the chip options.
- During option decoding, ADC module will be programmed to route the OPTION pin via AMUX and with an internal 100 K pull-up to the internal power rail of the PMIC.
- ADC reads the voltage at OPTION pin and decodes the option which is to be used in the PBS boot sequence and in the software.

NOTE: See the *PM6225/PM6375 Power Management IC Data Sheet* (80-27912-1) for the PM6225/PM6375 option pin details.

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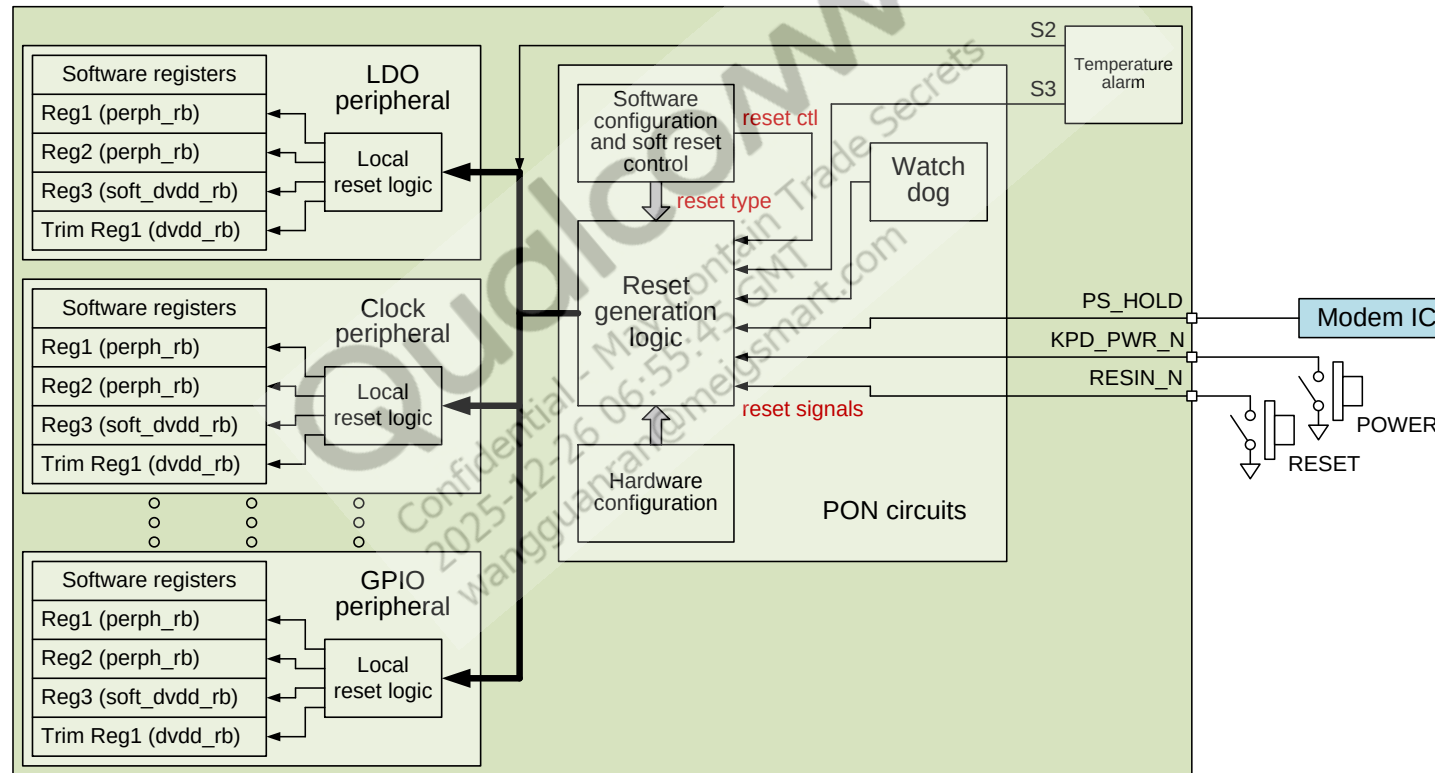
Resets

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Reset Architecture and Introduction

Multiple reset triggers are supported

- Examples: power-on button, overtemperature, PS_HOLD, and dedicated reset button
- By default, all peripherals follow all resets
- Software can individually configure each peripheral to ignore each reset trigger individually via masking



NOTE: Modem IC is applicable for SM6375 and SM6225 only.

Resets – Nomenclature

■ Reset triggers

- Internal and external triggers that are routed to the PMIC's power-on module; the module decides what action should be taken based on these triggers (stage 3 overtemperature, KPD_PWR_N, RESIN_N, and PS_HOLD)

■ Reset types

- Describes the behavior of the PMIC during a reset or shutdown event (like a hard reset event)
- The type of reset event is determined by configuration registers in the power-on module and broadcast throughout the PMIC using reset signals

■ Reset signals

- Signals that are generated in the power-on module and broadcast throughout the PMIC (for example, xVdd_rb, dVdd_rb, global_soft_rb, and shutdown1_rb).
- (Register) reset domains
- Several reset domains exist to allow some PMIC registers to maintain state throughout certain reset events

■ Reset stages

- Three stages of resets: software configurable bark, software configurable reset, and failsafe reset

Three-stage Reset and External Resets

■ Stage 1 reset: software-configurable bark

- The PMIC generates an interrupt, giving the SDM device the opportunity to fix the problem or gracefully reset the system. Example events that can cause a bark include:
 - Overtemperature indicates that the system is getting too hot.
 - The PMIC watchdog indicates that it has not kicked.

■ Stage 2: software-configurable bite

- If reset is ignored, the PMIC forces a reset event (selectable by software).

■ Stage 3: hardware mandatory bite

- The user can generate a mandatory reset by a long keypress of RESIN_N, KPD_PWR_N or RESIN_N plus KPD_PWR_N in combination.
- The stand alone or combination of reset triggers can also be selected as SBL by directly writing to the appropriate registers.
- Other comments:
 - Stage 3 cannot be disabled by software.
 - It resets the PMIC back to the factory default.
 - Stage 3 is a backup option if stage 1 and stage 2 fail.

■ External sources of reset (programmable)

- Power button (KPD_PWR_N)
- Reset button (RESIN_N)
- Power button + reset button
- Applications processor (PS_HOLD)
- Software write
- PMIC watchdog
- PMIC overtemperature sensor
- UVLO (mandatory immediate shutdown)

Reset Timers

#	Reset trigger	Stage 1 reset (bark) timer	Stage 2 reset (bite) timer	Stage 3 reset timer (note 3)
1	KPD_PWR_N (note 1)	0–10.256 sec	0–2 sec	0 sec 2–128 sec
2	RESIN_N (note 1)	0–10.256 sec	0–2 sec	0 sec 2–128 sec
3	KPD_PWR_N + RESIN_N (note 1)	0–10.256 sec	0–2 sec	0 sec 2–128 sec
4	PMIC watchdog (note 2)	0–127 sec	0–127 sec	–
5	PS_HOLD	–	–	–
6	Global reset	–	–	–
7	Overtemperature reset	–	–	–

NOTES:

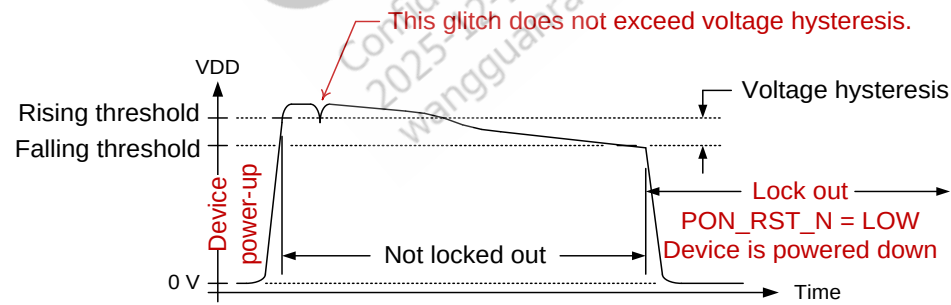
- See the [Reset Scheme](#) for configuration of these reset triggers in PM6225/PM6375.
- The default values of the debounce and the delay timers for the watchdog reset are 31 s and 32 s, respectively.
- Stage 3 timer is interdependent of stage 1 and stage 2.

UVLO

The UVLO circuit automatically turns off the PMIC at severely low VDD conditions. Although UVLO is a hardware feature, it allows for software interaction to realize additional features, such as SMPL recovery, power-on sequence abort, and watchdog timeout soft reset.

■ Operation

- At power-up, VDD must exceed a rising threshold for a sufficient debounce time to initiate the power-on sequence.
- Voltage hysteresis (175 mV default) and delays prevent minor glitches from being detected as UVLO events.
- If VDD drops below the falling threshold (UVLO rising threshold minus voltage hysteresis), a valid UVLO event is detected.
 - PON_RESET_N is cleared (LOW) and the device is powered down.
- The UVLO rising threshold voltage is programmable (1.675 V to 3.225 V in 50 mV increments).
 - Other than this programmable threshold, software is not involved in UVLO detection.
 - Hysteresis and time delays are not programmable, and UVLO events do not generate interrupts; they are reported to the modem IC via PON_RESET_N as part of power-down.



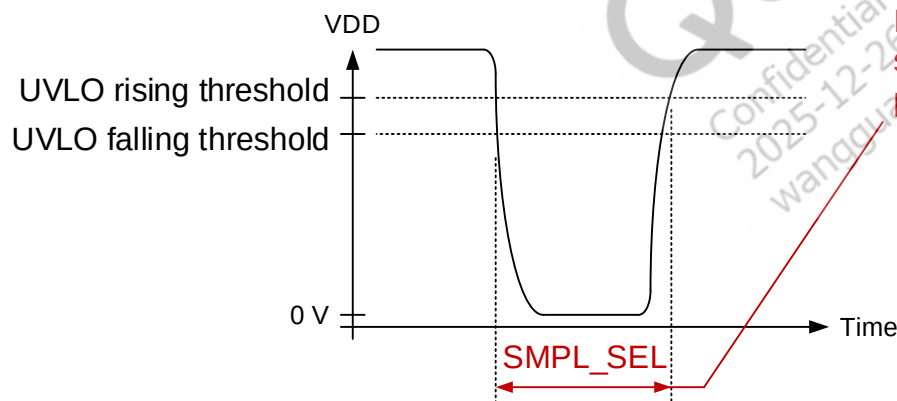
NOTE: See *PM6225/PM6375 Power Management IC Data Sheet (80-27912-1)* for UVLO value.

Overvoltage Lockout (OVLO)

- The OVLO setting is configured for 5.5 V during initial power-on on PM6225/PM6375.
- In order to power on successfully, VPH_PWR must be between the UVLO rising threshold and the OVLO falling threshold. These thresholds are not used after PON.
- In order for the device to remain on, VPH_PWR must remain between the UVLO falling threshold and the OVLO rising threshold. If these thresholds are exceeded, a fault will occur, and the chip will immediately shut down.
- UVLO and OVLO thresholds are set 150 mV lower and higher, respectively, on the slave PMICs, compared to the primary core PMIC. The intent is that the primary core PMIC should always be the one to detect, UVLO/OVLO event, and shut down the slave PMICs via FAULT_N. Therefore, UVLO/OVLO on slave PMICs is only a backup measure in a chipset environment.

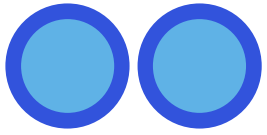
Sudden Momentary Power Loss (SMPL)

- When enabled by software, immediate and automatic recovery from a momentary PMIC power loss.
- If VDD drops out of range (< 2.475 V nominal), then it returns in-range within a programmable interval of between 0.5–2.0 s, and the recovery is executed.
 - Some operational details:
 - The UVLO event clears PON_RST_N; the PMIC is powered down.
 - A super capacitor or coin cell takes over as the SMPL power source.
 - If VDD returns to its valid range before timeout, a power-on sequence is initiated without software/user intervention, and an indication is sent to the modem IC to mention that the power was momentarily lost
 - If SMPL times out without VDD returning to its valid range, the handset must undergo a user-initiated power-on sequence whenever the next initializing event occurs.
 - SMPL operation must be enabled by software and requires a coin cell or keep-alive capacitor at VCOIN.
 - For a normal power-down, SMPL is disabled via PBS to avoid an inadvertent SMPL override.



If VDD recovers within this programmed interval a poweron sequence is immediately and automatically initiated by power management circuits without software intervention.

If RTC support is not needed when the battery is removed, a backup capacitor can be used on the VCOIN pin. A ceramic capacitor with an effective capacitance of 60 μ F can support SMPL for up to 2 s.



Section 3.4

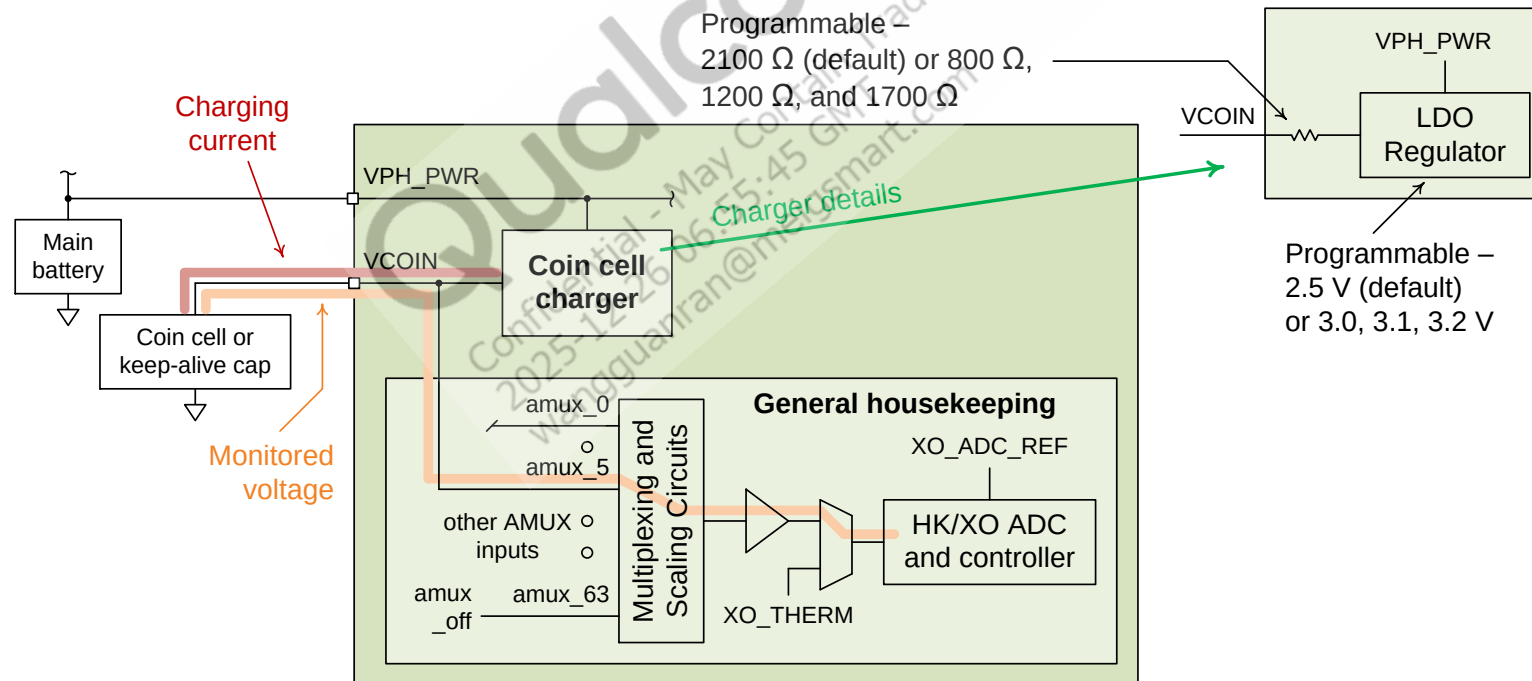
Coin Cell Support

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Coin Cell or Keep-alive Capacitor

Charging

- The programmed settings of the coin-cell charger remains valid even if the main battery is removed.
- If the coin-cell charger is enabled when the main battery is removed, charging will resume once the battery is reconnected.
- Charging resumes even if the phone is in an off state.
- Settings are reset to their default values only if the coin cell itself is removed.
- Coin-cell charger settings are not reset if an SMPL event occurs.



Coin Cell or Keep-alive Capacitor (cont.)

- The VCOIN of the PM6225/PM6375 device requires either a lithium manganese dioxide rechargeable coin cell or a keep-alive capacitor.
- The Panasonic ML-series (or an equivalent) is an example coin cell that complements the PMIC well.
- In addition to numerical specifications, consider the discharge characteristics – the coin-cell voltage should stay above 1.5 V for as long as possible, then drop off quickly from there.
 - This keeps the RTC, SMPS, and crystal oscillator circuits running during the off modes of the PMIC and ensures an abrupt turn-off as the backup voltage drops.
- When a coin cell is used
 - When the phone is powered off, the appropriate oscillator and real-time clock circuits continue to run.
 - These circuits are powered by the main battery, if present and valid.
 - If not, these circuits are powered by the coin cell.
 - A valid voltage on VCOIN is required to run the SMPL timer, but the coin cell may be replaced with an appropriate capacitor to maintain the SMPL feature only (as explained below).
- When a keep-alive capacitor is used
 - When the phone is powered off, the appropriate oscillator and real-time clock circuits continue to run.
 - These circuits are powered by the main battery, if present and valid.
 - If not, these circuits are powered down; the keep-alive capacitor does not store enough energy to power these circuits.
 - The keep-alive capacitor is sufficient to power the SMPL timer and supporting circuits only.

NOTE: The coin cell/keep-alive capacitor is not available on the PMI, and thus the PMI cannot maintain its volatile memory without a valid power source. RTC and SMPL are exclusively handled by the PM.



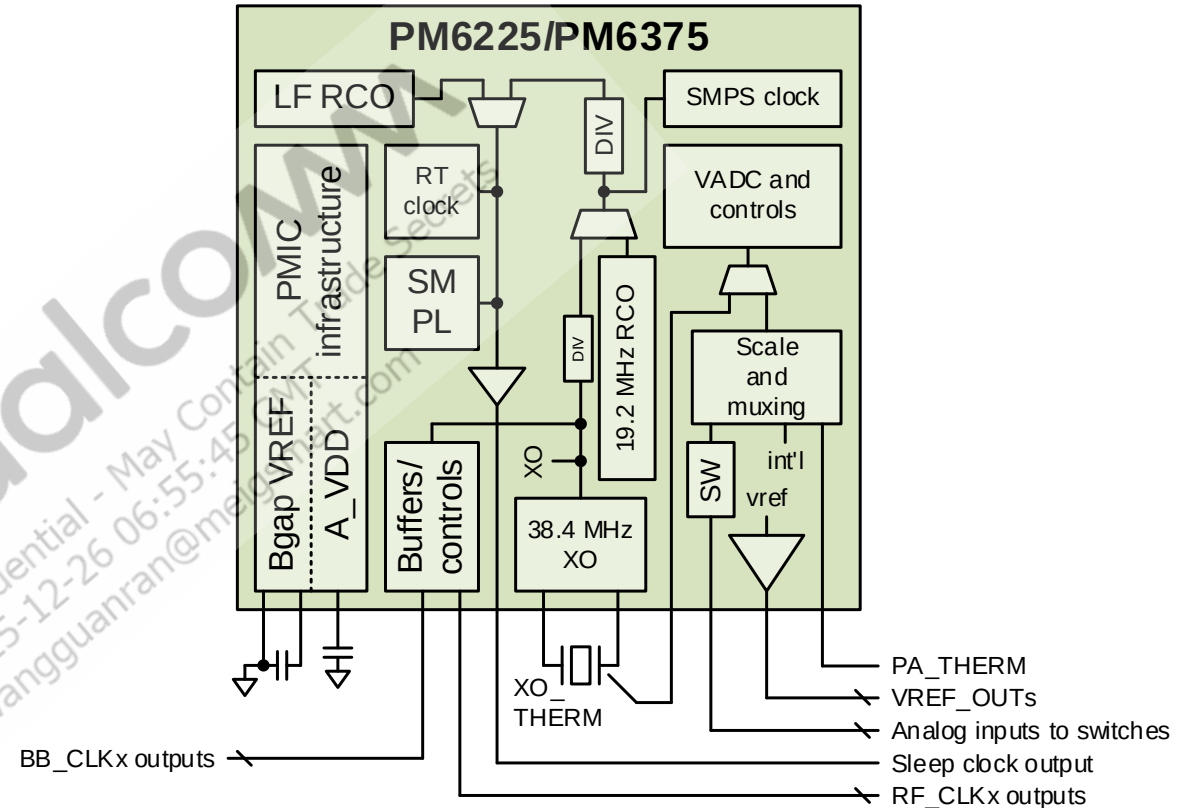
Section 4

General Housekeeping

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General Housekeeping

- Almost all general housekeeping functions are provided by the PM6225/PM6375 device.
- General housekeeping includes:
 - Analog multiplexer (AMUX) and ADC
 - Clocks
 - 38.4 MHz XO
 - Sleep clock
 - Other on-chip clock functions
 - System-wide clock distribution
 - Overtemperature protection
 - IC infrastructure





Section 5

Clocks

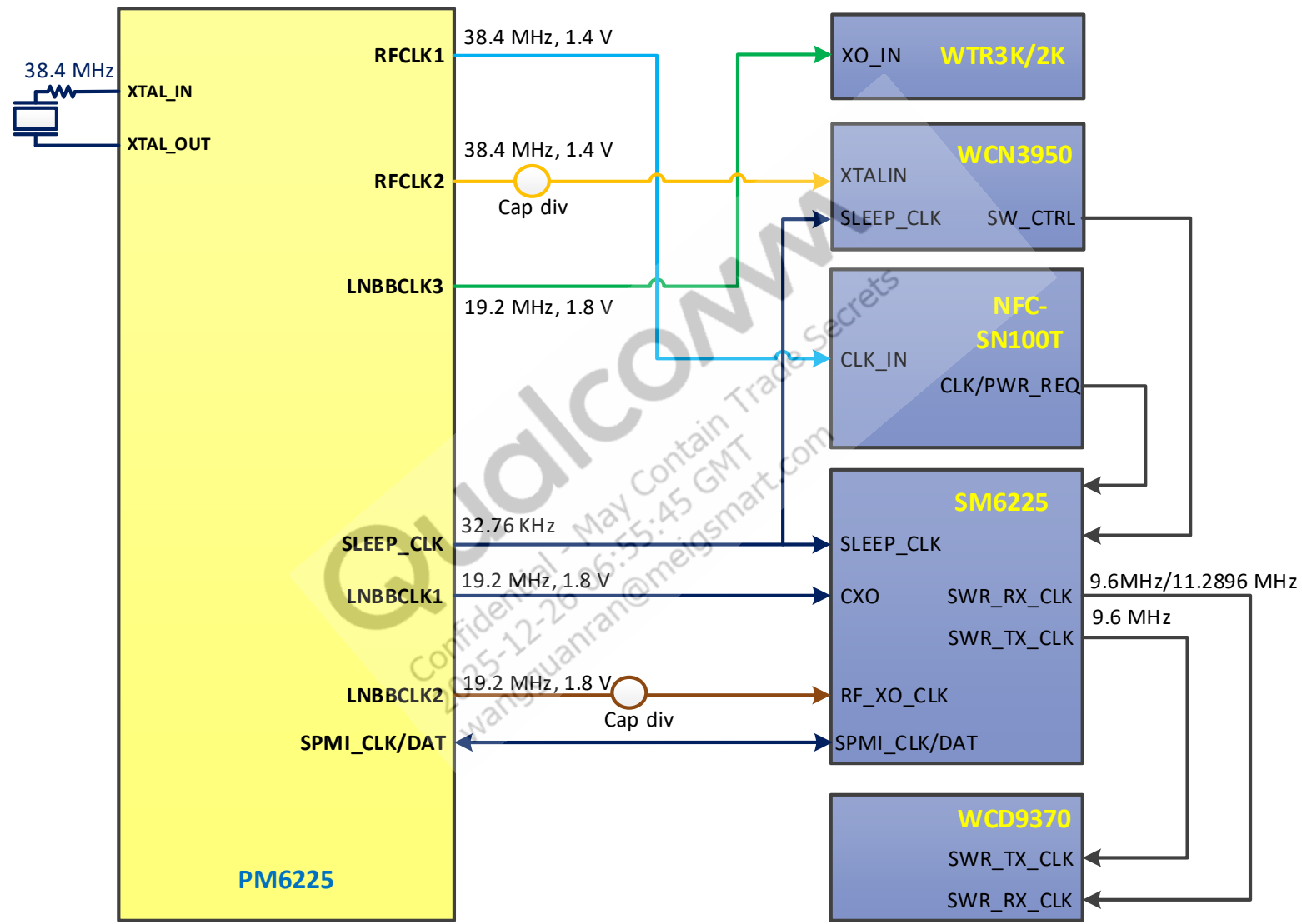
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PM6225 Clocks

Several PM6225 device clocks and clock outputs are used for general housekeeping functions and elsewhere throughout the system.

- 38.4 MHz crystal oscillator (XO) circuit, the master clock of the system.
- Several system clock controller/buffer circuits
 - Two for RF circuits (RFCLK1 and RFCLK2 for NFC and WCN)
 - Three low-noise baseband clocks (LNBBCLK1 for the SM6225 digital system CXO input, LNBBCLK2 for WLAN, and LNBBCLK3 for WTR or other uses)
- Divided and buffered clocks support internal functions (SMPS, sampling, and so on).
- Divided and buffered clocks are available at certain GPIO ports.
- 19.2 MHz RC oscillator for power-up and emergency backup.
- Calibrated low-frequency RF oscillator.
- The sleep clock is divided down from the XO or uses a calibrated low-frequency RC oscillator (also supports the real-time clock (RTC)).

PM6225 Clocks (cont.)





Section 6

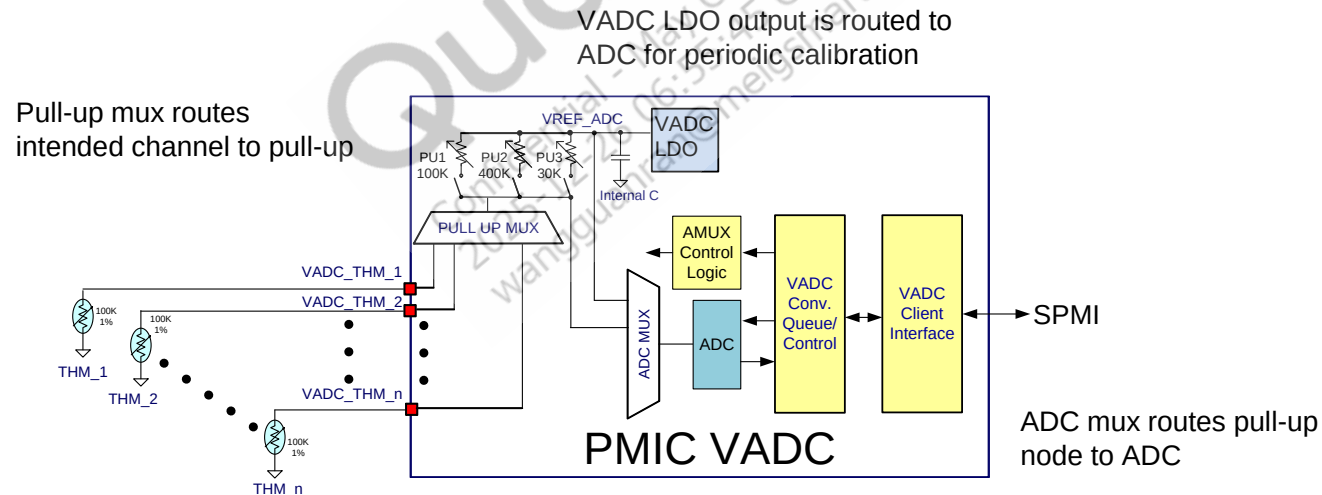
Internal AMUX Pull-up

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Internal Pull-up Resistors

Optimizing cost and performance, QTI PMICs feature internal pull-up resistors

- Three pull-up values (30 K, 100 K, 400 K)
- Trimmed to within 0.5% accuracy: better tolerance than discrete R
- Low temperature coefficient (< 100 ppm/C): better than discrete R
- Bias source is a calibrated internal LDO: no package pin
- LDO is automatically controlled by ADC hardware
- Each thermistor is only biased when measured: saves power





Section 7

Configurable I/Os

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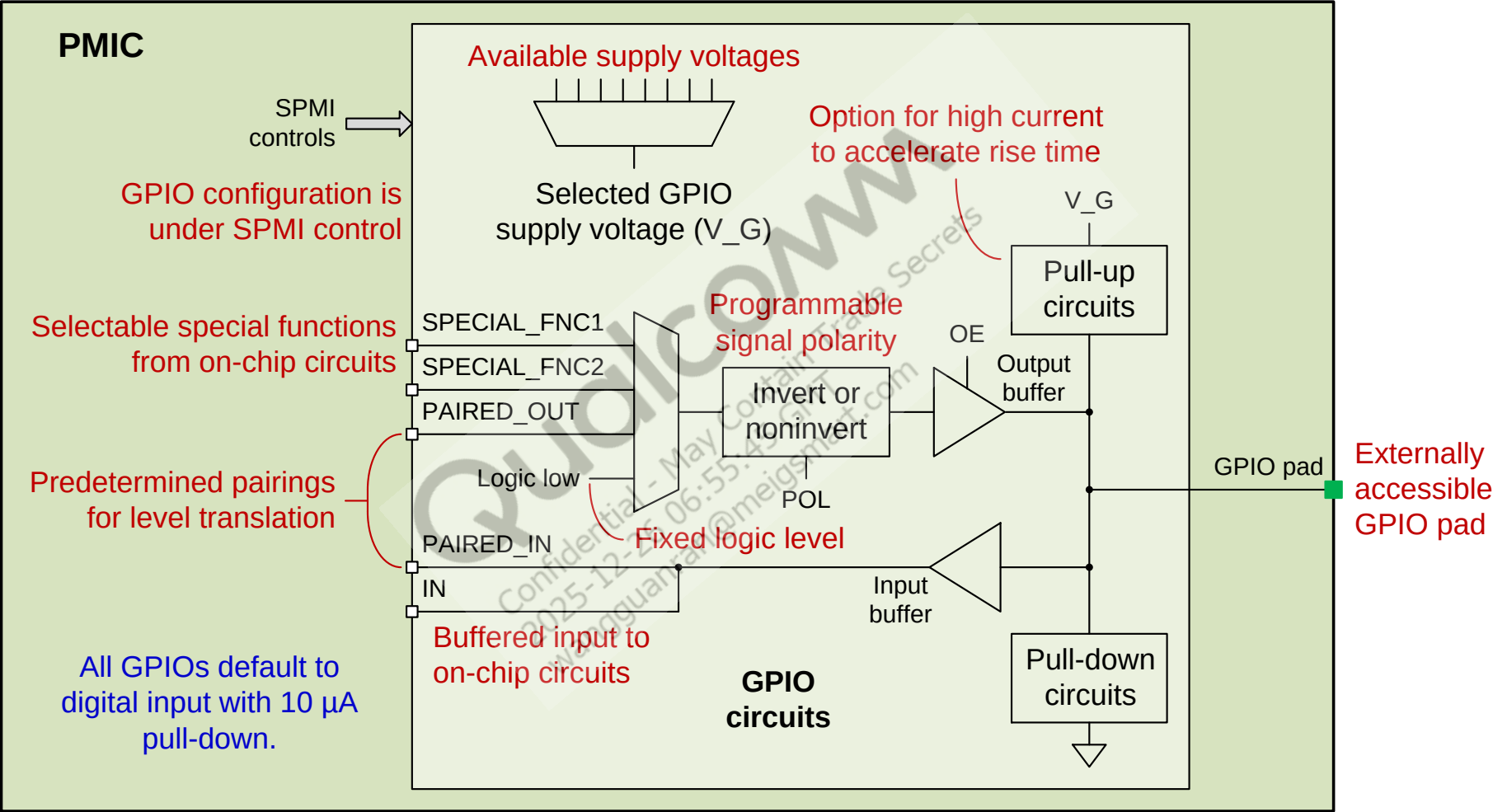
PMIC GPIO Pins

- The PM6225/PM6375 device has nine GPIOs
 - GPIO_1 to GPIO_5 are LV GPIOs (1.8 V only)
 - GPIO_6 to GPIO_9 are MV GPIOs (VPH_PWR or 1.8 V)
- Selectable I/O supply voltages are listed below.
- PM6225/PM6375 device generic features – can be implemented by any GPIO.

Power source	Default	GPIO availability
VDD_MSM_IO	1.800 V	GPIO_01 to GPIO_09
VPH_PWR	3.600 V (typical)	GPIO_06 to GPIO_09

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PMIC GPIO Pin Diagram



GPIO Configurations

See the *PM6225/PM6375 Power Management IC Data Sheet* (80-27912-1) for the GPIO assignments.

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PM6225/PM6375 Unused Pin Terminations

Because PM6225/PM6375 is the core PMIC, there are not many unused functions or pins. The pins that are optional and unused can be terminated in the following ways:

- Unused LDOs
 - Default ON LDOs still require a load capacitor
 - Default OFF LDOs can be left no connect (NC)
 - VDD inputs should still be connected to power even if unused
- Unused clock outputs can be left NC
- Unused GPIOs can be left NC
- Unused PON sequencing pins, such as CBL_PWR_N, KPD_PWR_N, and so on, can be left NC.



Section 8

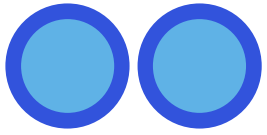
PM6375 and PM6225 Layout Guidelines

8.1 PM6375 Layout Guidelines

[70](#)

8.2 PM6225 Layout Guidelines

[91](#)

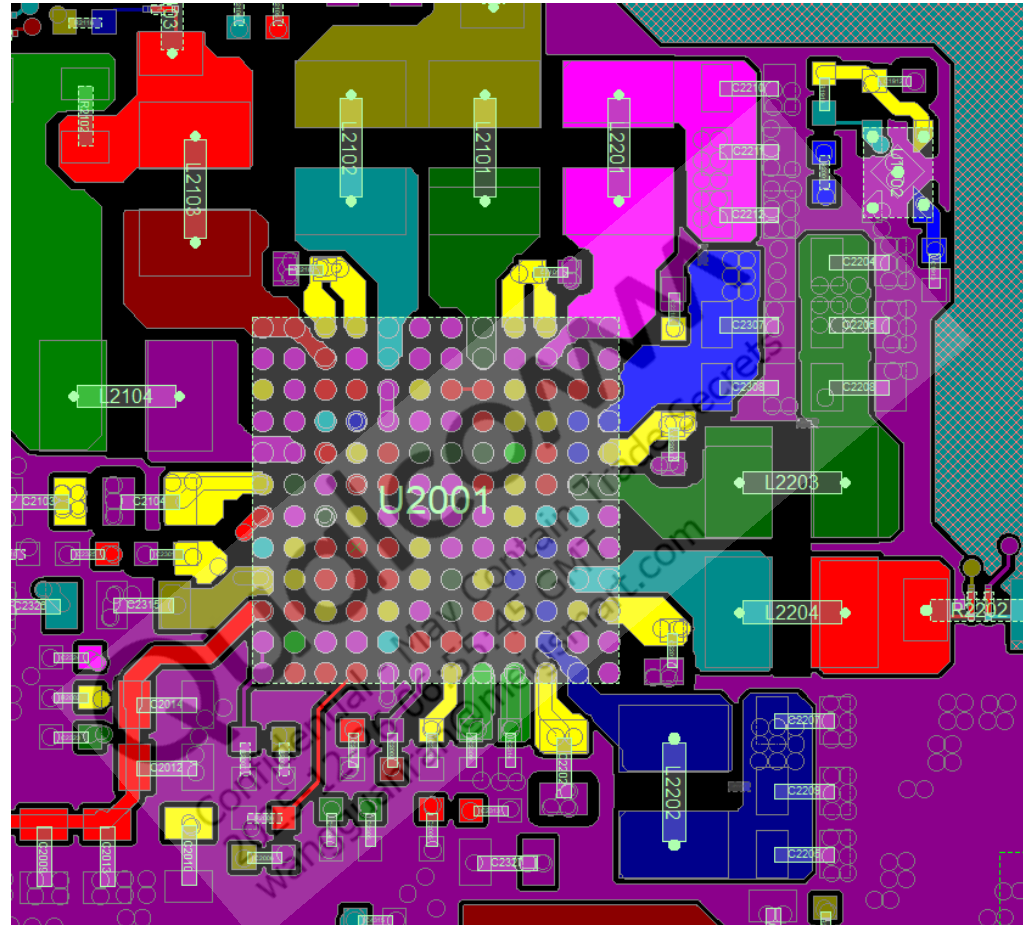


Section 8.1

PM6375 Layout Guidelines

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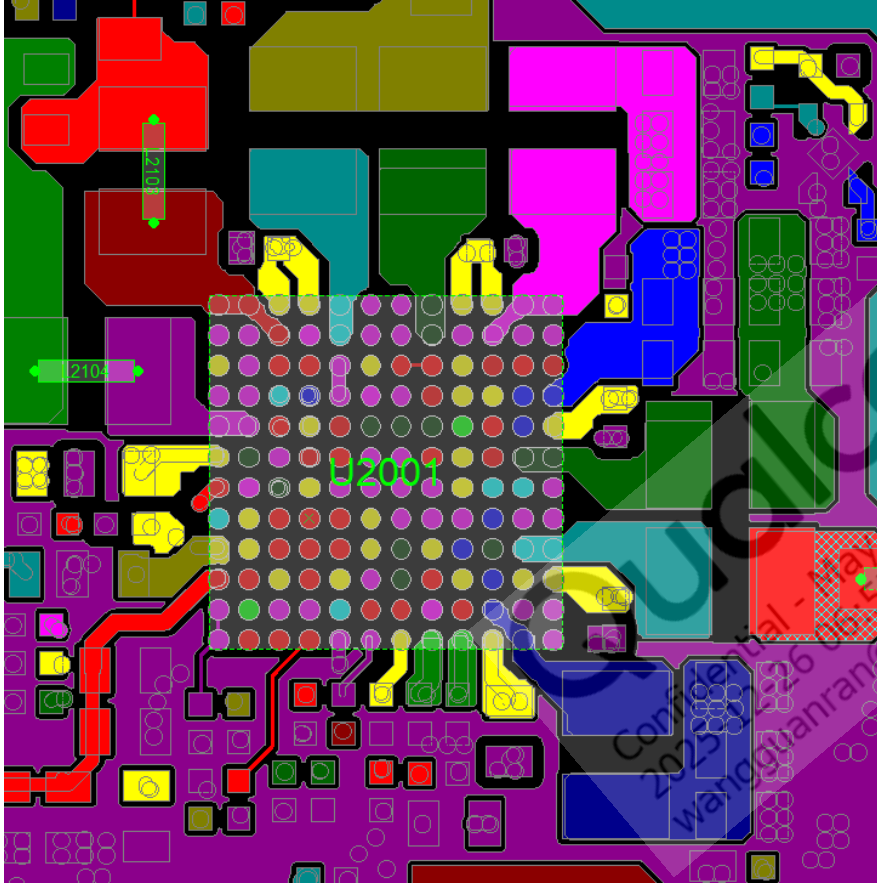
BOM Placement



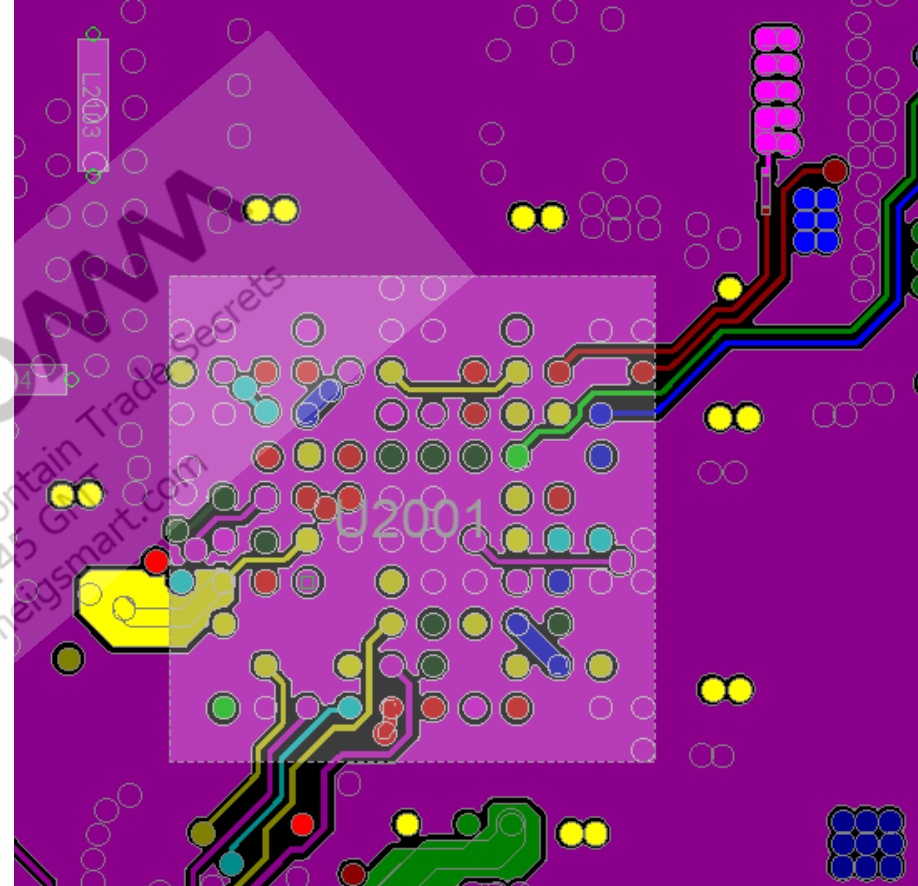
NOTES:

- The components shown here provide general guidance on BOM placement.
- The component number indicators shown in the figure do not have any reference to any schematic and are only used to indicate the type of component (that is, R = resistor, C = capacitor, and so on).

Routing Example

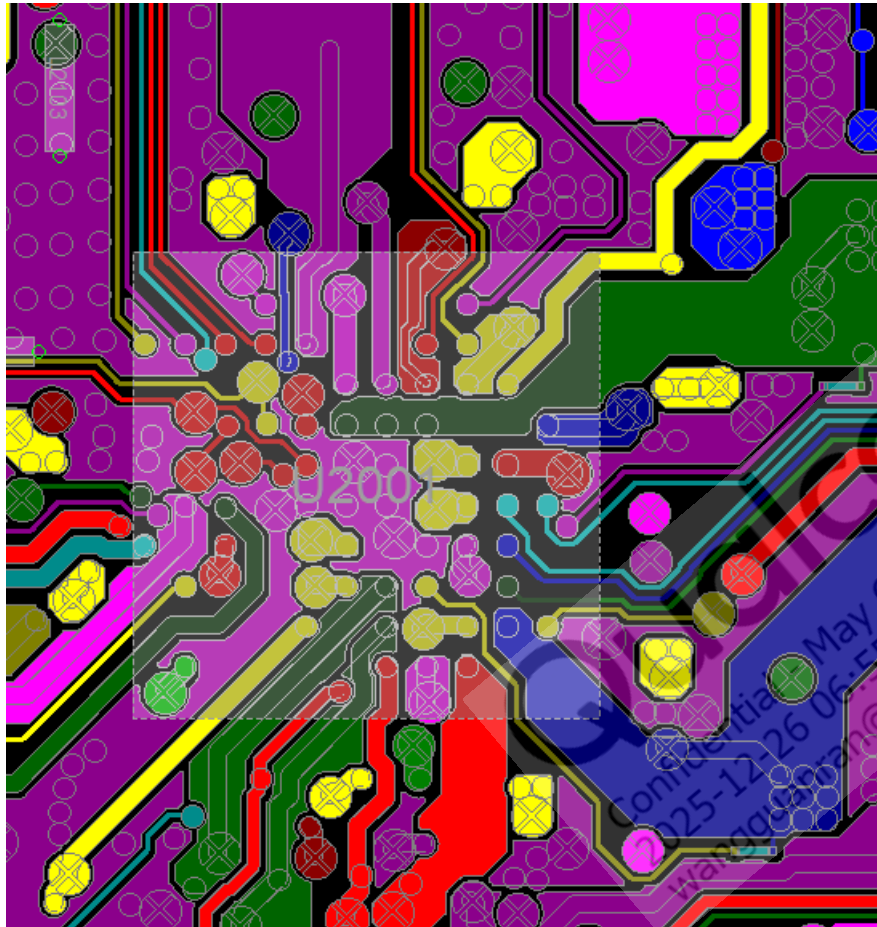


Layer 10 (PM6375 - bottom layer)

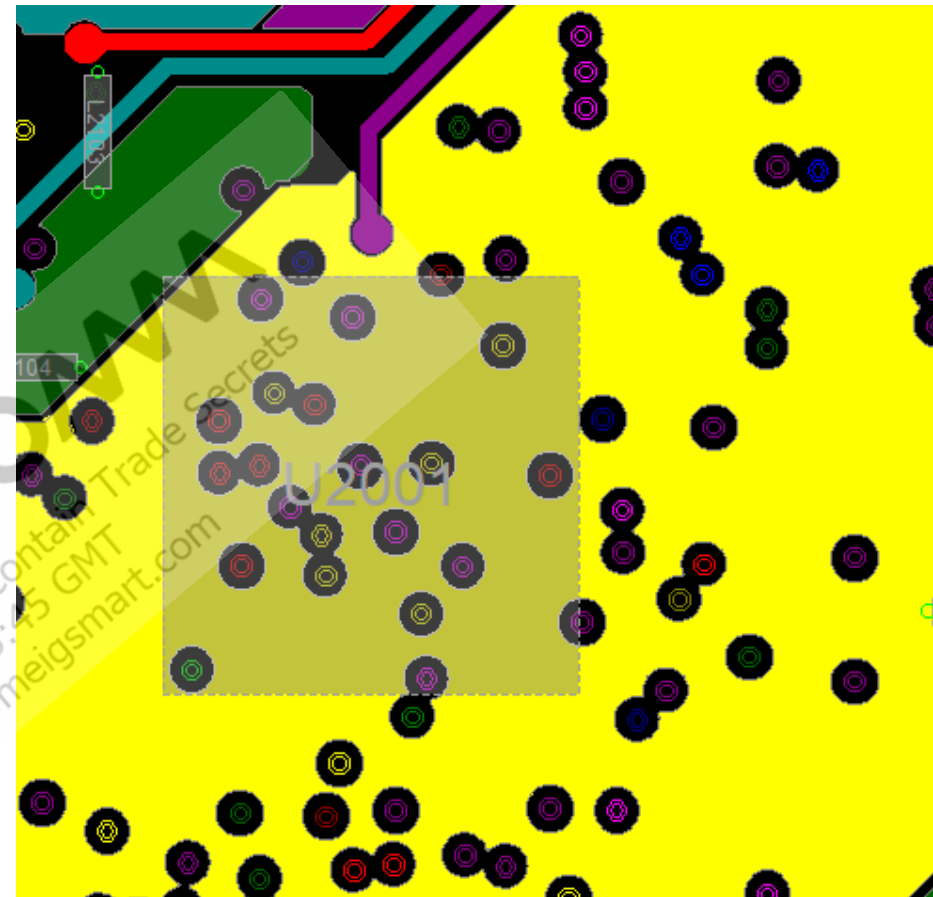


Layer 9 (PMIC GND)

Routing Example (cont.)



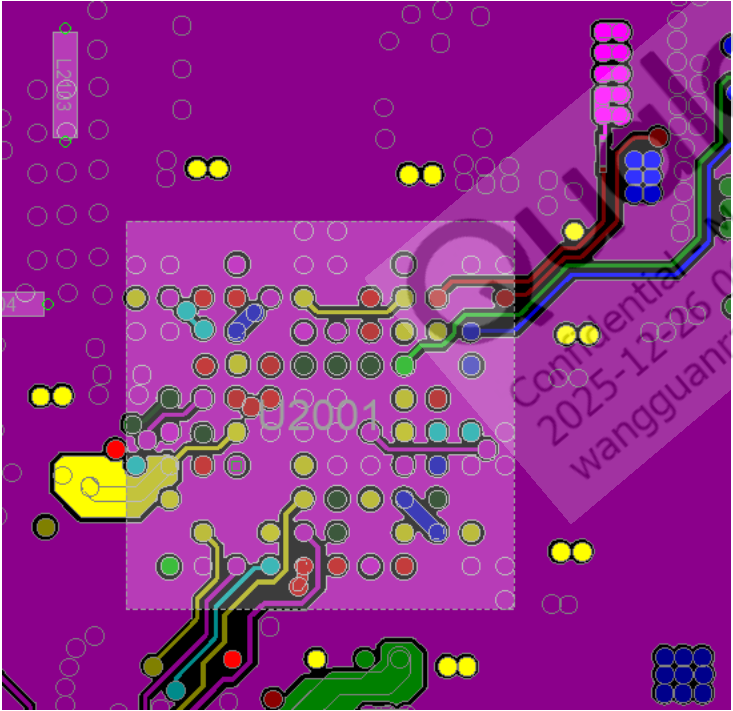
Layer 8 (Breakout/routing layer)



Layer 7 (VPH layer)

GND

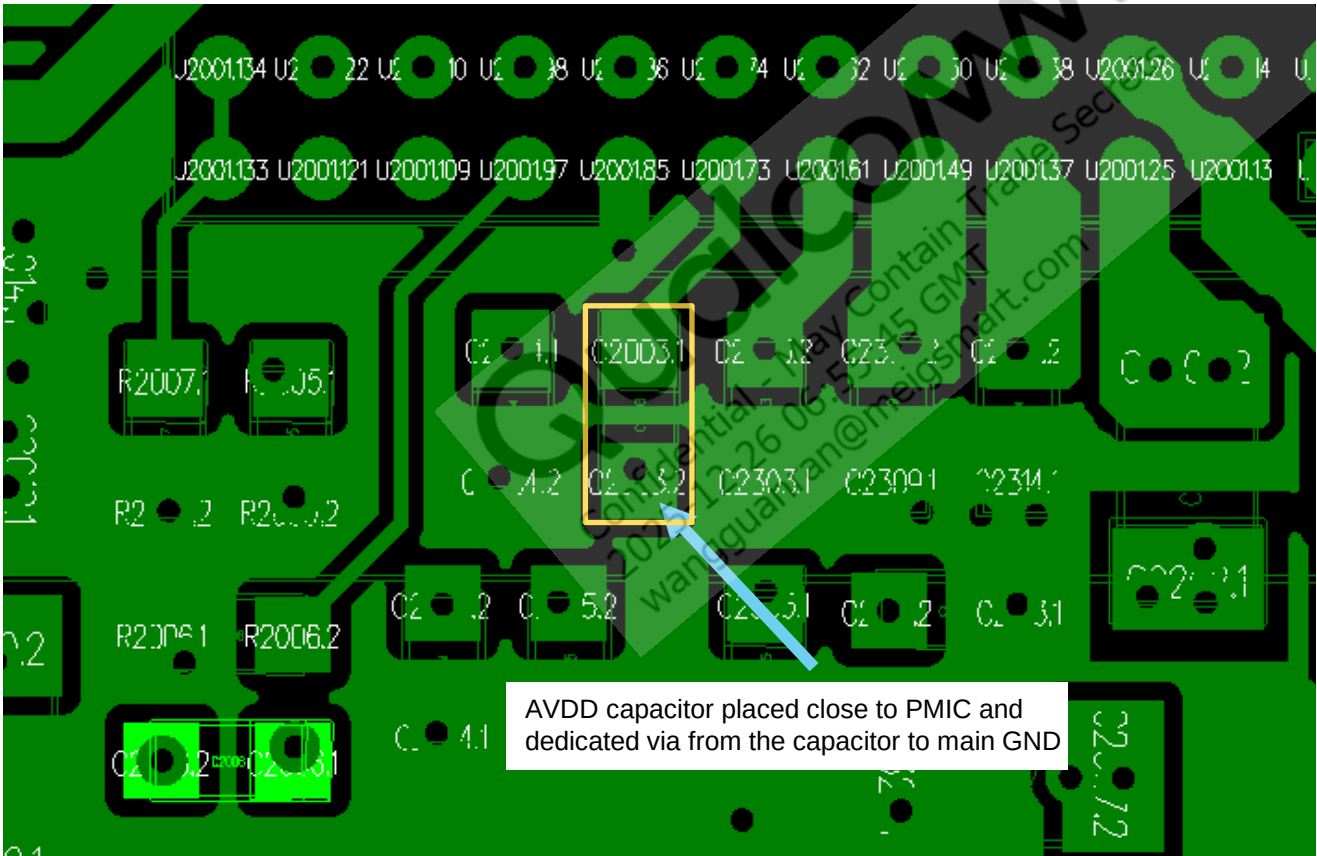
Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	GND	Cutouts are permitted under only LDO/INFRA modules. The non-GND μ Vias spacing must meet this requirement to avoid cutouts. Area directly under PMIC must be a solid ground plane. It can only be pierced by μ Vias. This is, henceforth called PMIC GND.	0.34	Ball pitch	–	mm	If cutouts occur anywhere else, insufficient shielding of noisy traces may occur facilitating noise propagation. Impedance specifications in other rules may risk being violated.
High	GND	The PMIC GND plane must be stitched to main GND plane of system with specified μ Via spacing.	0.35	0.75	1	mm	Non-stitched GND act as antennas and propagate noise. Without stitching, SMPS noise may be shuffled to CMN_GND balls.



L9 - PMIC GND

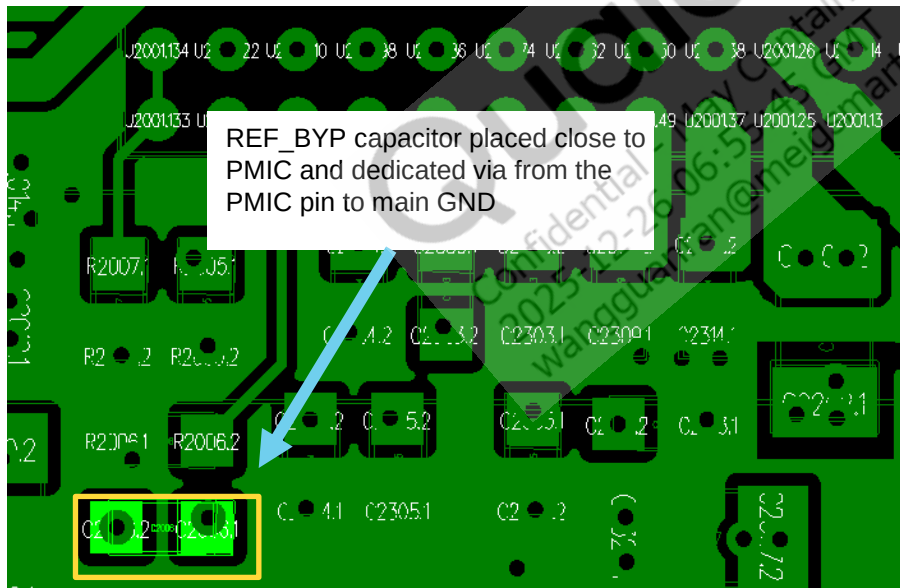
AVDD_BYP

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	AVDD_BYP	AVDD_BYP cap must be placed beyond the inductors. Add μ Via to GND at cap pad. Loop impedance from AVDD_BYP pin to CMN_GND pin with AVDD_BYP cap shorted must meet the requirement.	—	50	100	m Ω	AVDD supplies various analog circuits in the PMIC, and any noise picked up on this trace can cause many issues.
				2	10	nH	



REF_BYP/REF_GND

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	REF_BYP/ REF_GND	Place REF_BYP cap in a quiet area. Have a dedicated trace from REF_GND pin to GND pad of REF_BYP cap. ONLY add μ Via to main GND plane at REF_GND pin. Follow the rule above unless the Reference Design does otherwise. Loop impedance from REF_BYP pin to REF_GND pin with REF_BYP cap shorted must meet the requirement.	—	50	200	m Ω	This is the reference voltage to the entire PMIC. Any noise that couples on to either signal will show up at all regulator outputs. Via at PMIC pin minimizes buck switching HF noise picked up by REF_GND. This HF noise coupling can cause a DC shift in MBG output.
				----- 2	----- 10	----- nH	



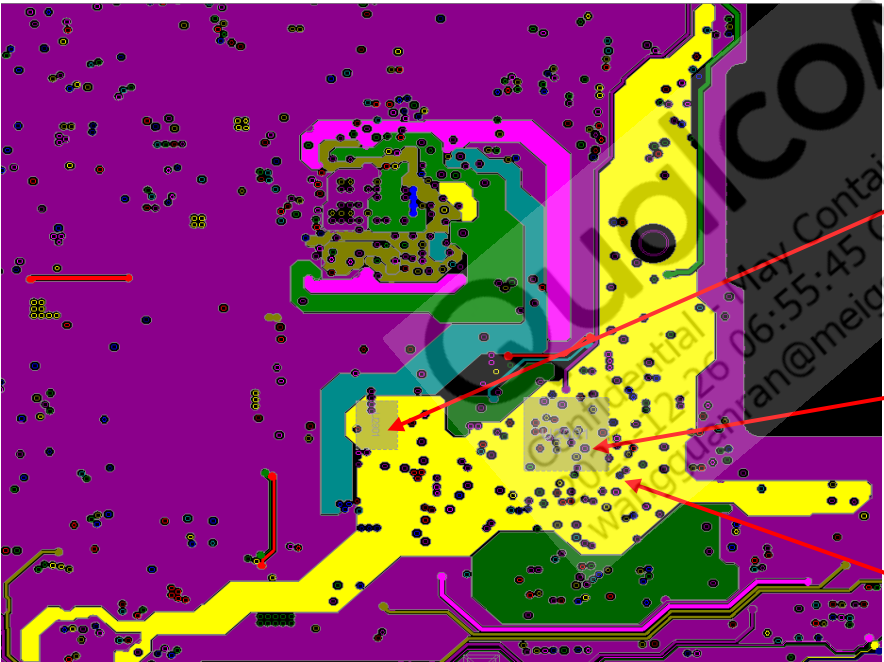
ADC

Priority	Net	Routing requirement	Target			Units
			Min	Typ	Max	
High	AMUXx/GPIOs	Coupling to ADC inputs (AMUX pins or GPIOs) from any aggressor must meet the requirement.	0	40	50	fF

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VPH Plane

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VPH	VPH plane should be a solid plane under the PMIC, extending to VSW pads of inductors. Cutouts are permissible but signal routing is prohibited. The non-VPH μ Vias spacing must meet this requirement to avoid cutouts.	0.3	Ball pitch	10	mm	VPH must be a low impedance plane in order to meet the SMPS Zxx/Zxy requirement. Violating this requirement will lead to adding more SMPS Cin BOM to meet the Zxx/Zxy requirement.
Medium	VPH	The VPH plane must be within specified number of layers from the PMIC GND.	1	1	2	Layers	Forms a distributed capacitance with the PMIC GND plane, thereby foregoing the need for buck input capacitance. Without it, voltage droop on VPH and cross talk between the bucks may occur.



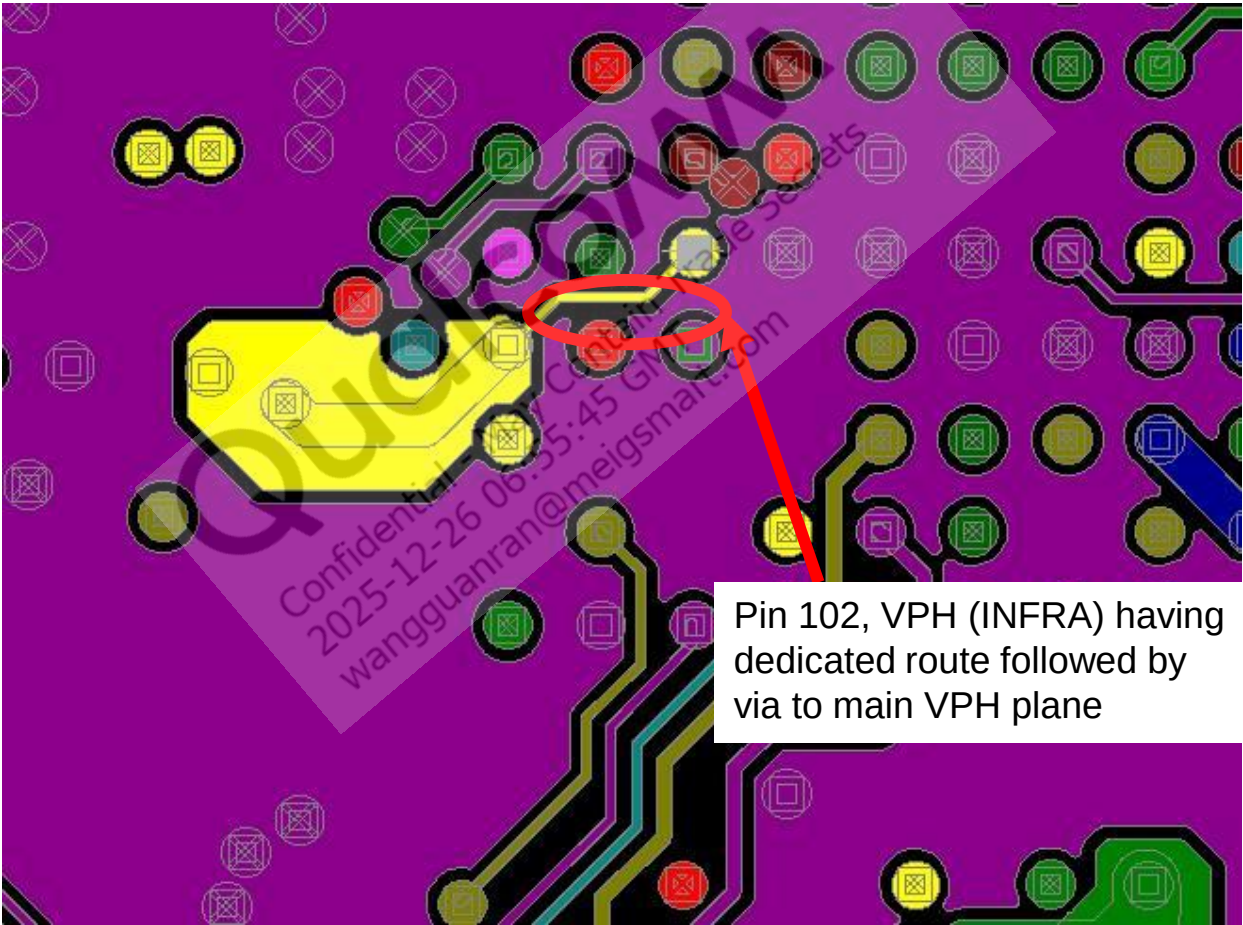
PMR735A

PM6375

L7 – VPH flood

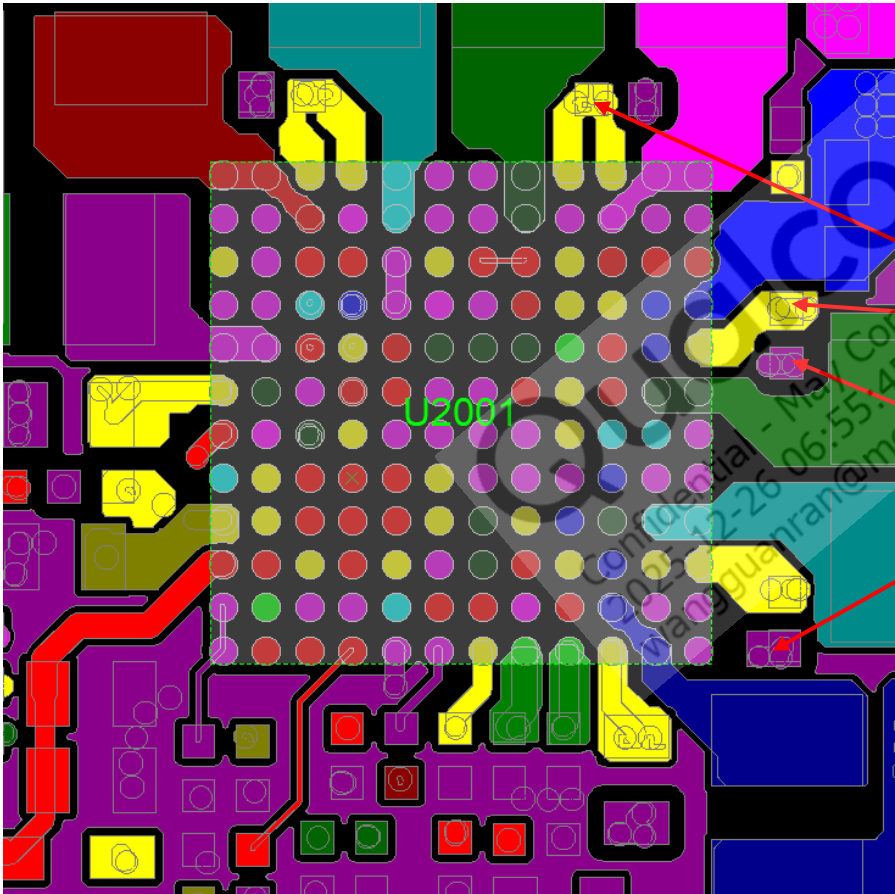
VPH Plane (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VPH(CMN)	The Zxy specifications against all the SMPS VSW and VPH. The cross impedance, Zxy, should meet the impedance specifications. Zxy is calculated with all capacitors populated and batfet open.	–	3 ----- 100	5 ----- 180	mΩ ----- pH	This is the reference voltage to the entire PMIC. Any noise that couples on to either signal will show up at all regulator outputs.



VPH Plane (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VPH Cap/GND	μ Vias to the PMIC GND are placed at the capacitor pads.	1	2	2	μ Vias	Violating this requirement may cause violation of the Zxx/Zxy requirement thereby coupling noise to other parts of the component layer.
High	VPH	Loop DCR from the buck VDD_Sx to GND_Sx, with battery terminals shorted and batfet shorted.	0	10	20	m Ω	High resistance will decrease the efficiency and affect buck operation.

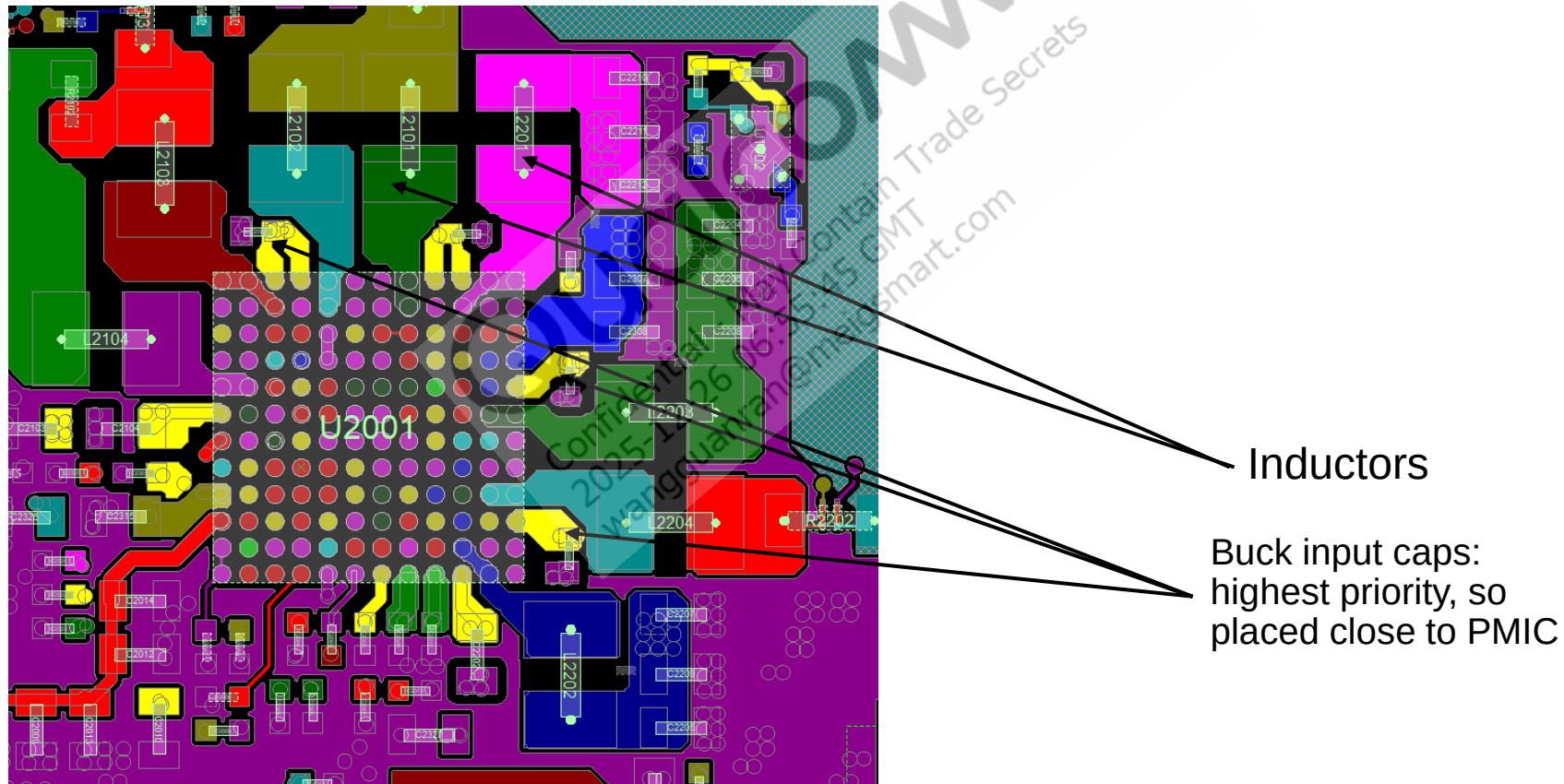


VPH μ Vias to VPH plane

μ Vias to PMIC GND placed at VPH cap pads

Component Placement Priority

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	VREG_Sx/ VREG_Lx/ INFRA	Passive component placement priority from closest to furthest from PMIC: 1) Buck Cin 2) Inductors 3) nLDO input/output caps 4) SMPS Cout for local sense 5) Caps for PMIC infrastructure 6) SMPS Cout for remote sense	—	—	—	—	The impedance requirements may not be met. Unnecessary noise propagation from aggressors such as VSW and VDD_Sx



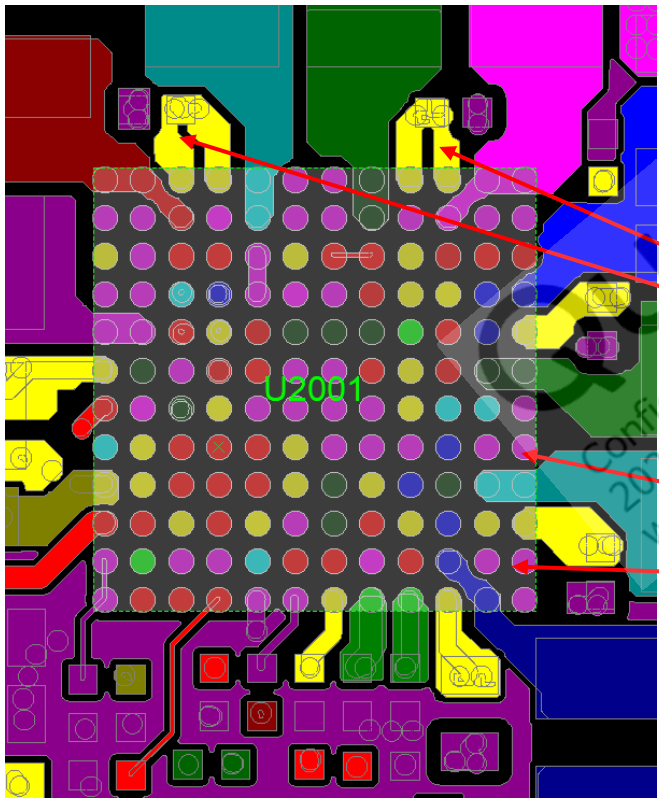
VCOIN

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Low	VCOIN	VCOIN trace must meet the DCR requirement.	–	100	200	mΩ	Too higher resistance will reduce the effectiveness of VCOIN when UVLO is hit. SMPL time window reduces (sudden momentary power loss)

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Buck Input

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VDD_Sx	Option 1: If VDD_Sx pins see a cap directly, route a direct trace to cap and have two μ Via to VPH plane located at capacitor pads. Option 2: If two buck inputs see one input capacitor, create a slot in the VPH trace to ensure the cross impedance is reduced.	–	–	–	–	Insufficient μ Vias to VPH plane will increase Zxx.
High	GND_Sx	Each GND_Sx pin must have μ Via to layer 2 PMIC GND plane.	1	1	1	μ Via/pin	Noise may couple to other traces if these vias are located off-chip.



Slot in VPH trace for shared caps in bottom layer

Every GND_Sx pin has a μ Via to PMIC GND

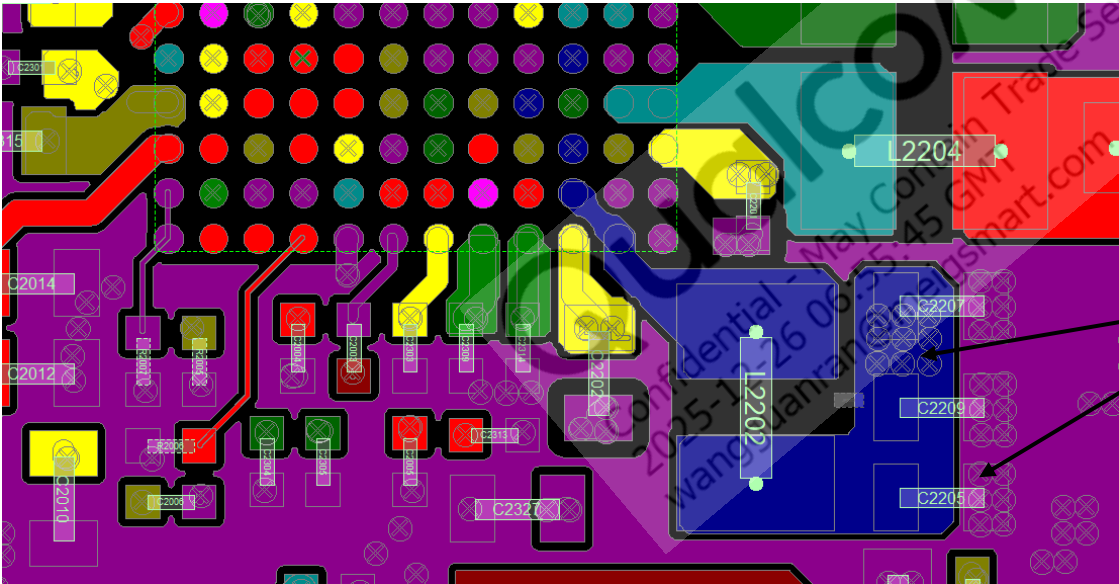
Buck Input (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VDD_Sx/ GND_Sx	The self impedance of each buck input, Zxx, should meet the impedance specifications. Loop impedance from VDD_Sx pins to GND_Sx pins with VPH capacitors modelled.	See SM6375 Chipset Power Delivery Network Specification document (80-26086-1P) for target values			mΩ ----- pH	1. Voltage spikes resulting in damage to, or reduced lifetime of PMIC. 2. EMI due to radiation caused by high transient currents in the VIN_SX/PGND traces.
High	VDD_Sx/ GND_Sx	To minimize the influence of buck over another, the cross impedance, Zxy, should meet the impedance specifications. Zxy is calculated with all capacitors populated and batfet open.				mΩ ----- pH	Reduced buck efficiency and performance due to mistriggering of buck comparators.

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Buck Output

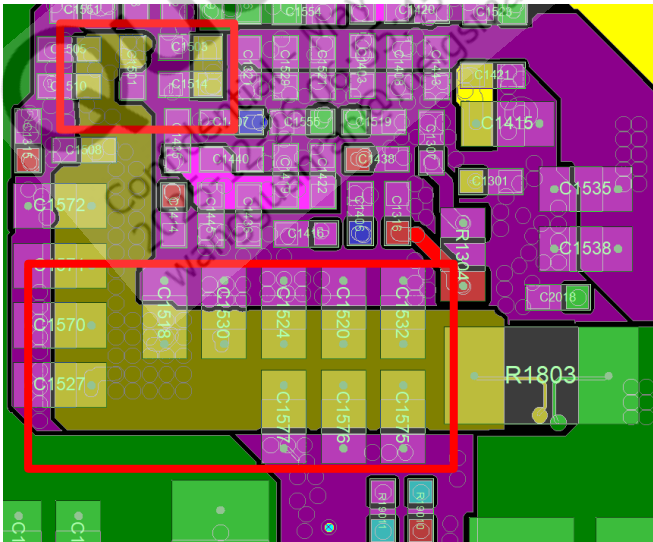
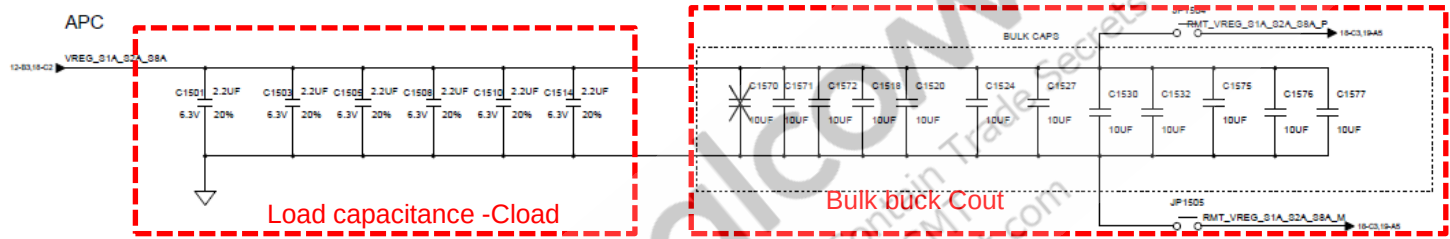
Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	VREG_Sx output	Resistance from VSW pins to FB point with inductor shorted should meet the target for each of the buck individually. For multiphase bucks short all VSW pin to equipotential.	0	12	20	mΩ/phase	Reduced buck efficiency due to high PCB IR loss.
Medium	Buck Cout	All power routes must be star-routed from buck Cout. Use sufficient μVias at Cout pads to handle I _{max} . Power route for LDO input may be tapped before buck Cout in some cases. Contact QTI if there is a need for this.	2	3	5	μVia/pad	Taps before Cout will have higher noise and worse regulation.



μVias at Cout to handle I_{max} for subregulation

Buck Output (cont.)

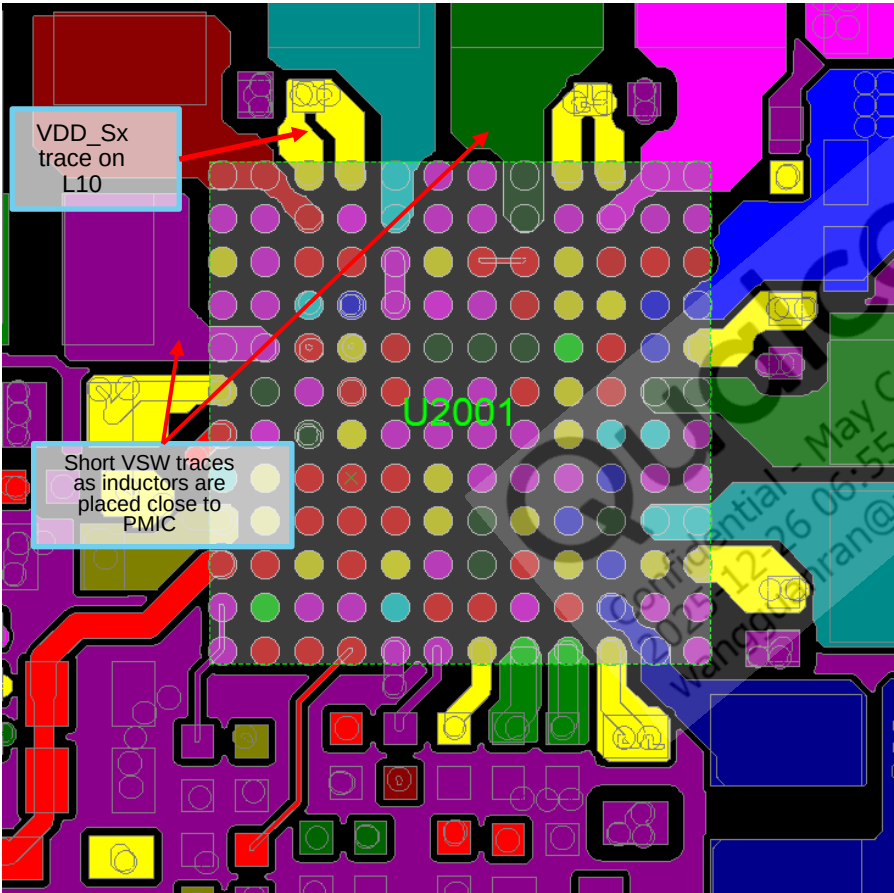
Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	Buck Cout	Impedance between any bulk buck Cout capacitors.	0	0.05 ----- 100	5 ----- 500	mΩ ----- pH	Buck Cout should not be too far apart from each other, otherwise this may lead to buck instability and reduced efficiency.
High	FTS specific Cout + Cload	Total load capacitance distributed across all the loads must not exceed the requirement (n = number of phases).	0	4 × n	10 × n	μF	Buck instability
High	FTS specific Cout + Cload	Ensure the impedance to any load capacitor from bulk buck Cout meets requirement, where the ratio of total Cout per phase and total Cload. (x = Cout per ph/Cload total).	0	0.1x ----- 0.25x	0.75x ----- 1x	mΩ ----- nH	Buck instability



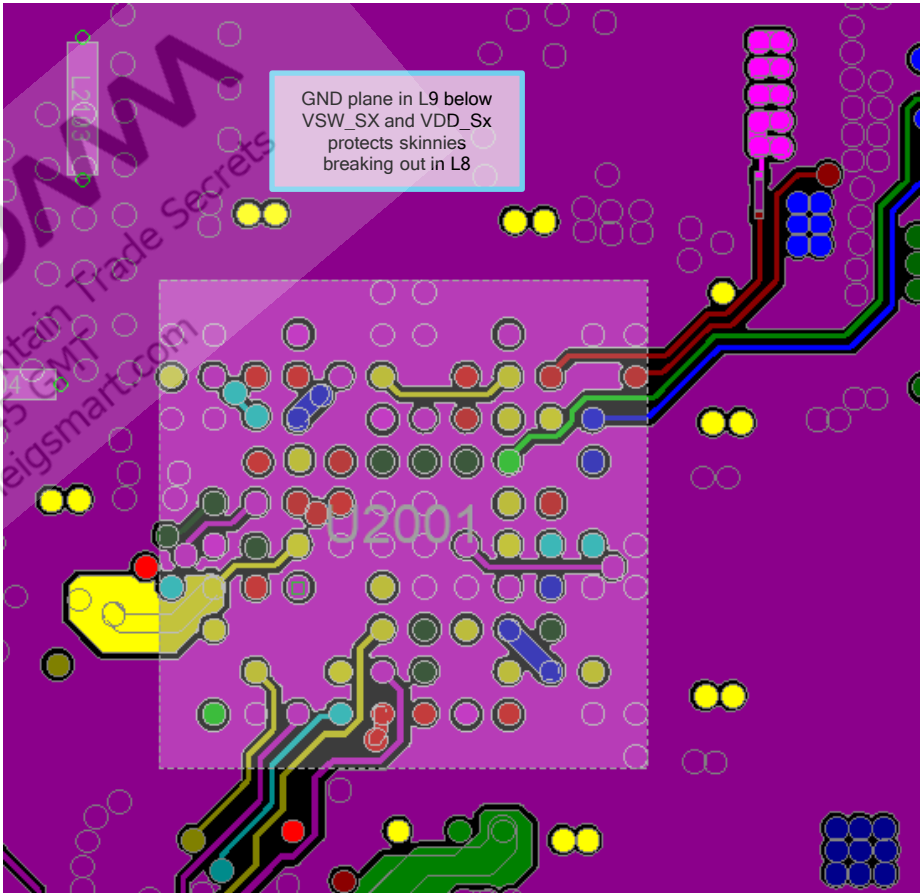
Buck Cout's placed in close vicinity to meet the impedance targets

Buck VSW_Sx/VDD_Sx

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VSW_Sx/ VDD_Sx	Very noisy traces. They are strong aggressors. Any non-GND capacitive coupling to these traces should be kept within requirement.	0	0.5	1	fF	Sensitive signals in proximity will be impacted.



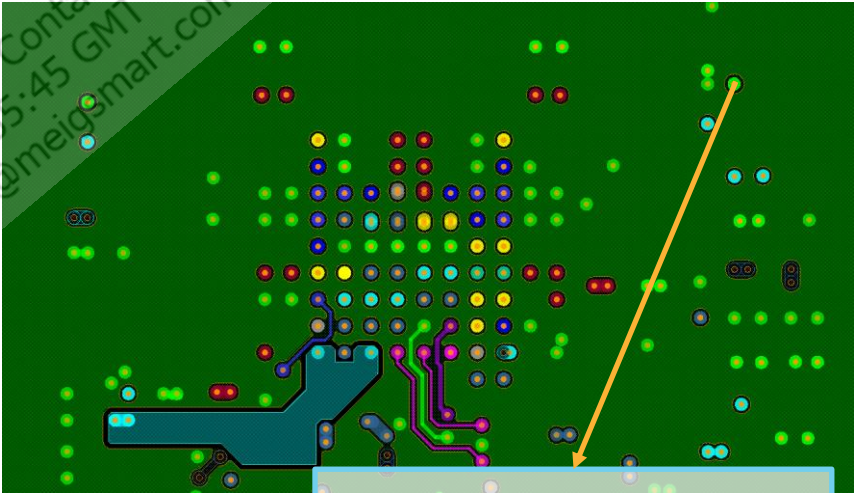
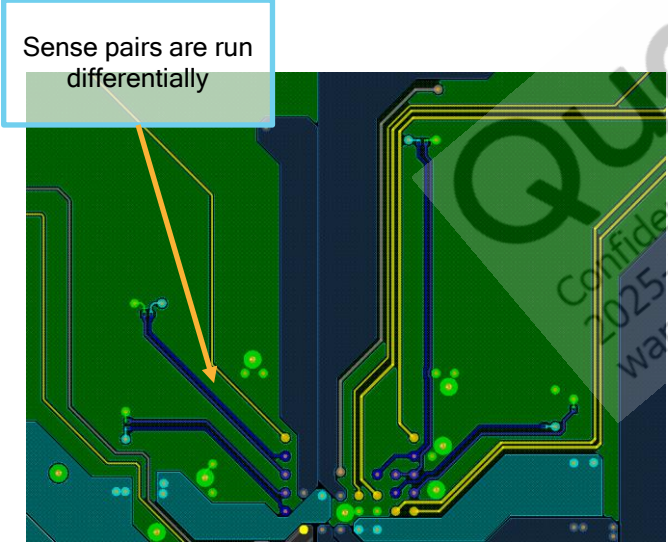
L10



L9 GND

Buck Feedback

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_Sx_SNS and RMT_GND_SNS	FTS sense pairs must be routed differentially. Capacitive coupling between each other should meet the specification. For multiphase domains, short sense pins at PMIC and only one sense pair should be broken out.	50	100	10000	fF	Reduced buck efficiency due to mistriggering of buck comparators.
Medium	VREG_Sx_SNS and RMT_GND_SNS	Tap SNS pair at or downstream of Cout with a dedicated, noncurrent carrying via. Resistance between SNS tap point and Cout (lumped) must meet the DCR requirements.	—	<1	3	mΩ	Taps before Cout will have higher noise and worse regulation and elevated output ripple.
Medium	HFS with remote FB	Impedance between inductor and remote FB point need to meet the impedance requirements.	—	1 ----- 0.1	33 ----- 5	% of DCR of inductor (mΩ) ----- % of Inductance of Inductor (pH)	Buck instability and reduced efficiency. Every 10 mΩ degrades efficiency by 0.6%.



LDO Input and Output

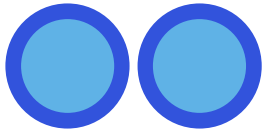
Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_Lx	All LDO PDN requirements for input and output routes must meet requirements listed in the respective chipset specific -1P document.	0	45–95	100	% of spec	Degraded LDO performance and functionality, for example, instability, degraded PSRR, and poor voltage regulation.

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VIO_OUT

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VIO_OUT	Loop impedance from VIO_OUT pin to its output capacitor must meet the requirement.	–	40	100	mΩ	
				2	10	nH	

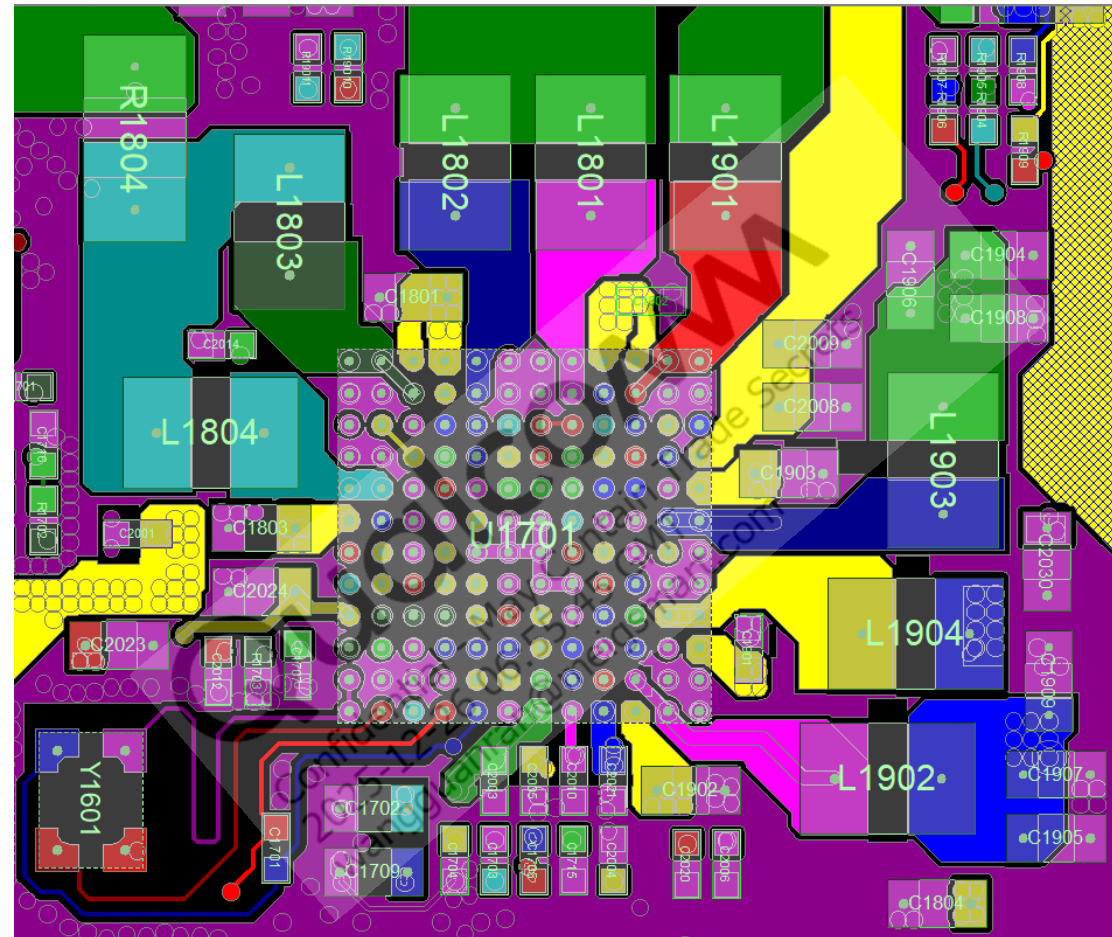
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Section 8.2

PM6225 Layout Guidelines

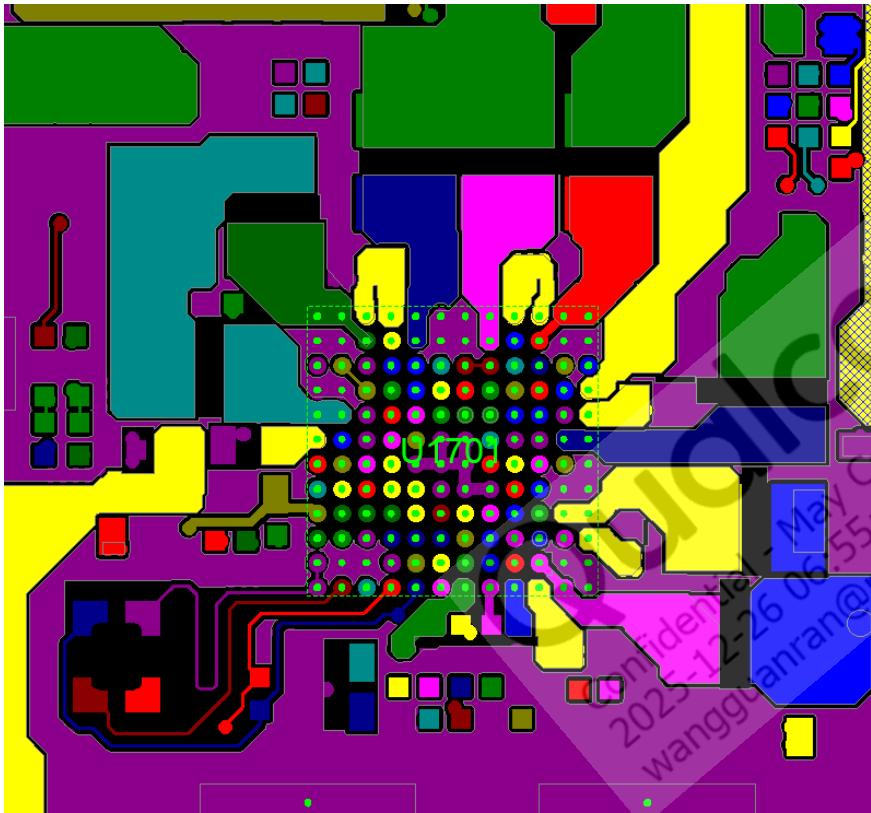
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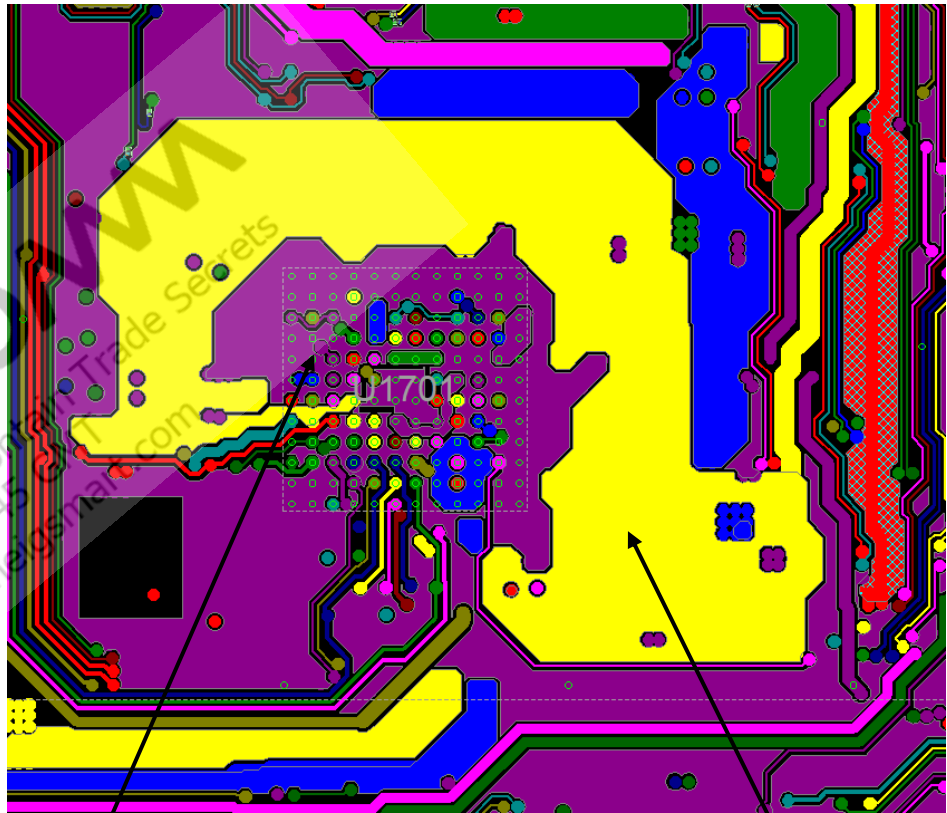
NOTES:

- The components shown here provide general guidance on BOM placement.
- The component number indicators shown in the figure do not have any reference to any schematic and are only used to indicate the type of component (that is, R = resistor, C = capacitor, and so on).

Routing Example



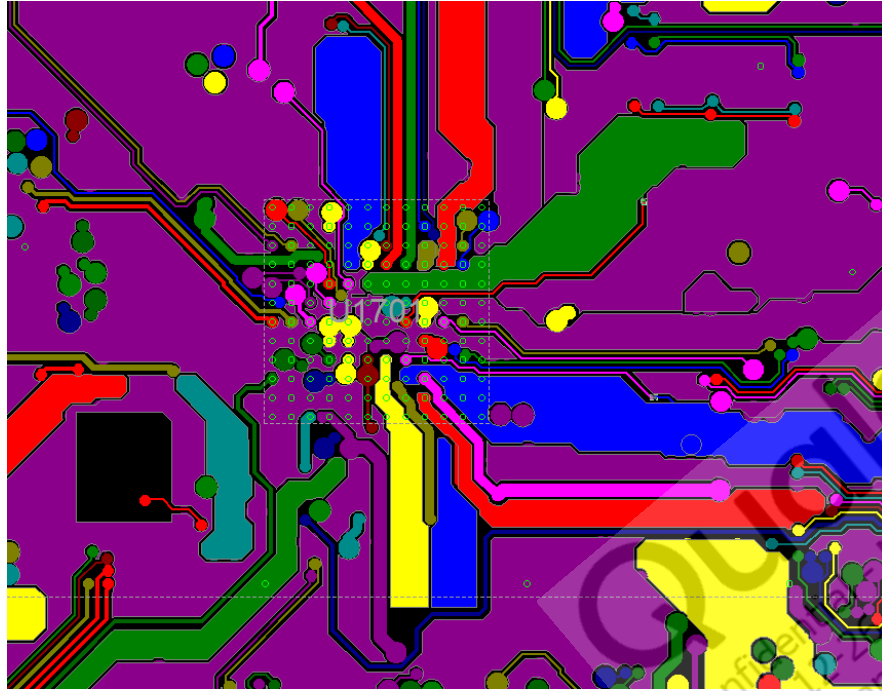
Layer 8 (PM6225 placement bottom layer)



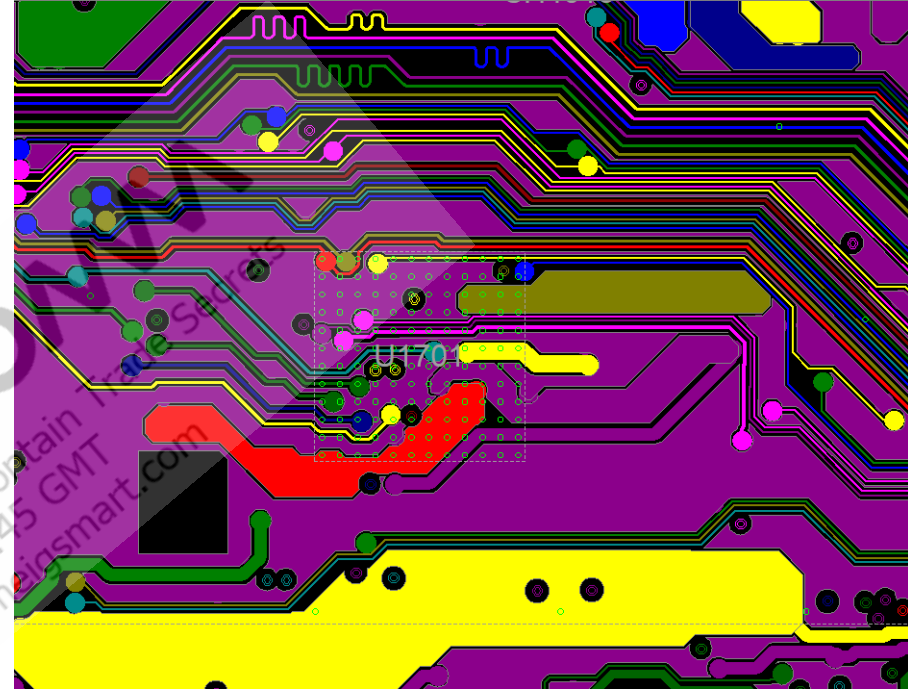
Layer 7 (PMIC GND + VPH)

VPH

Routing Example (cont.)



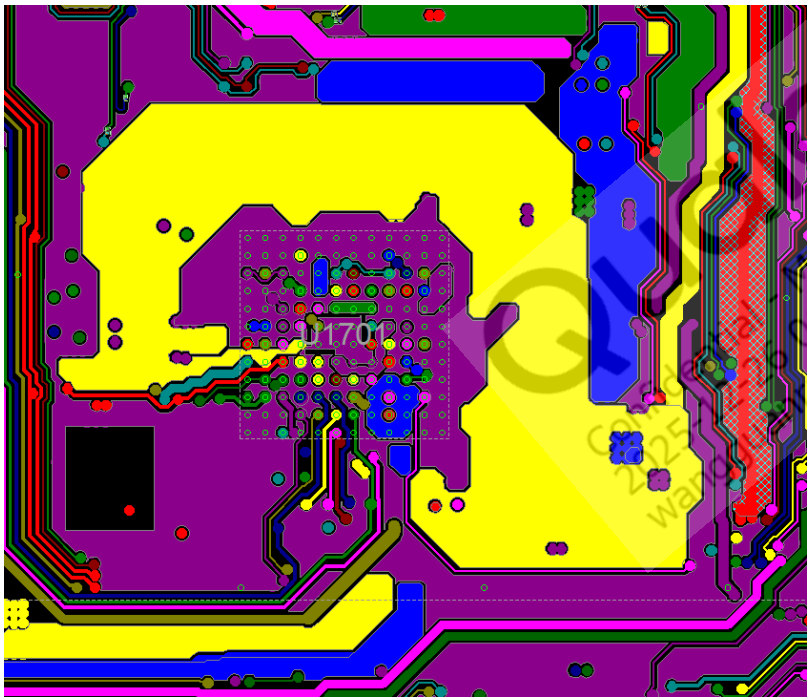
Layer 6 (Breakout/routing layer)



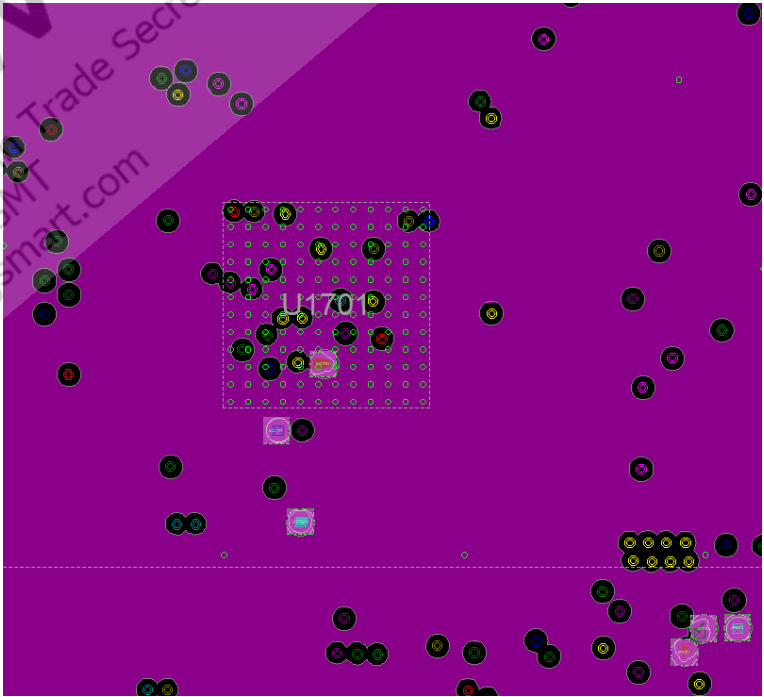
Layer 5 (Breakout/routing layer)

GND

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	GND	PMIC GND plane must be stitched to main GND plane of system with specified μ Via spacing.	0.35	0.75	1	mm	Non-stitched GND act as antennas and propagate noise. Without stitching, SMPS noise may be shuffled to CMN_GND balls.



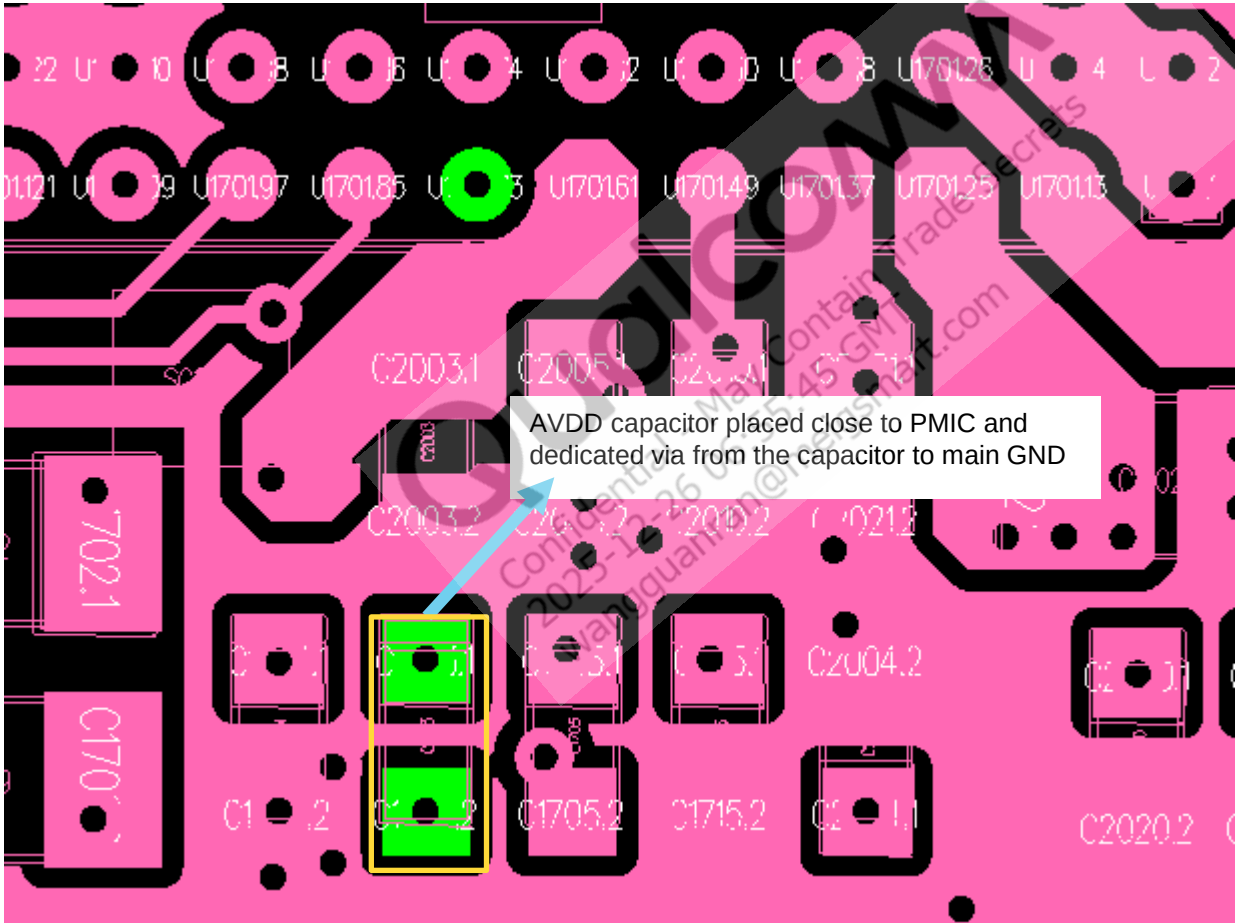
L7 - PMIC GND + VPH



L4 - Main GND

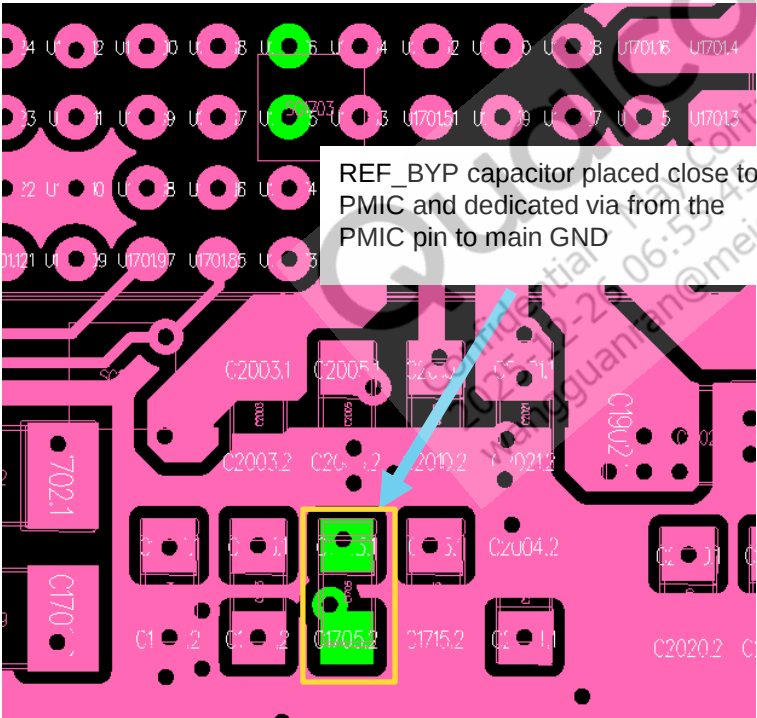
AVDD_BYP

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	AVDD_BYP	AVDD_BYP cap must be placed beyond the inductors. Add μ Via to GND at cap pad. Loop impedance from AVDD_BYP pin to CMN_GND pin with AVDD_BYP cap shorted must meet the requirement.	–	50 ----- 2	100 ----- 10	m Ω ----- nH	AVDD supplies various analog circuits in the PMIC, and any noise picked up on this trace can cause many issues.



REF_BYP/REF_GND

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	REF_BYP/ REF_GND	Place REF_BYP capacitor in a quiet area. Have a dedicated trace from REF_GND pin to GND pad of REF_BYP capacitor. ONLY add μVia to main GND plane at REF_GND pin. Follow the rule above unless Reference Design does otherwise. Loop impedance from REF_BYP pin to REF_GND pin with REF_BYP cap shorted must meet the requirement.	–	50 ----- 2	200 ----- 10	m Ω ----- nH	This is the reference voltage to the entire PMIC. Any noise that couples on to either signal will show up at ALL regulator outputs. Via at PMIC pin minimizes buck switching HF noise picked up by REF_GND. This HF noise coupling can cause a DC shift in MBG output.



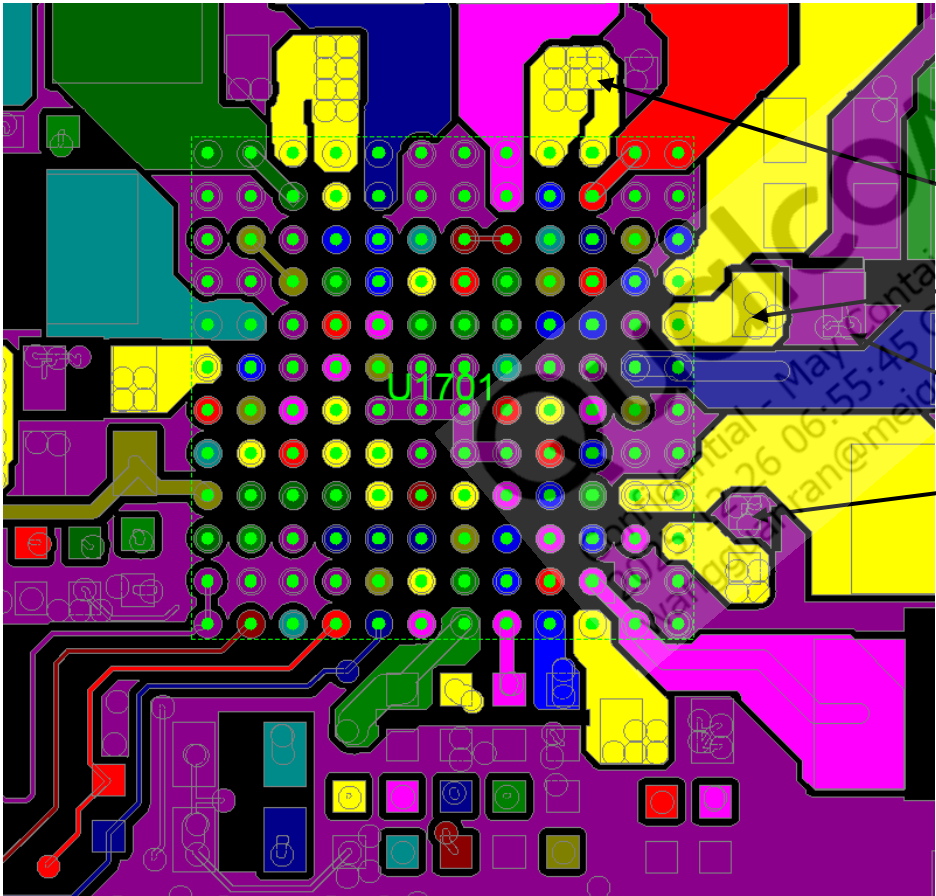
ADC

Priority	Net	Routing requirement	Target			Units
			Min	Typ	Max	
High	AMUXx/GPIOs	Coupling to ADC inputs (AMUX pins or GPIOs) from any aggressor must meet the requirement.	0	40	50	fF

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VPH Plane

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VPH cap/GND	μ Vias to PMIC GND are placed at capacitor pads.	1	2	2	μ Vias	Violating this requirement may cause violation of the Zxx/Zxy requirement thereby coupling noise to other parts of the component layer.
High	VPH	Loop DCR from buck VDD_Sx to GND_Sx, with battery terminals shorted and batfet shorted.	0	10	20	m Ω	High resistance will decrease efficiency and affect buck operation.

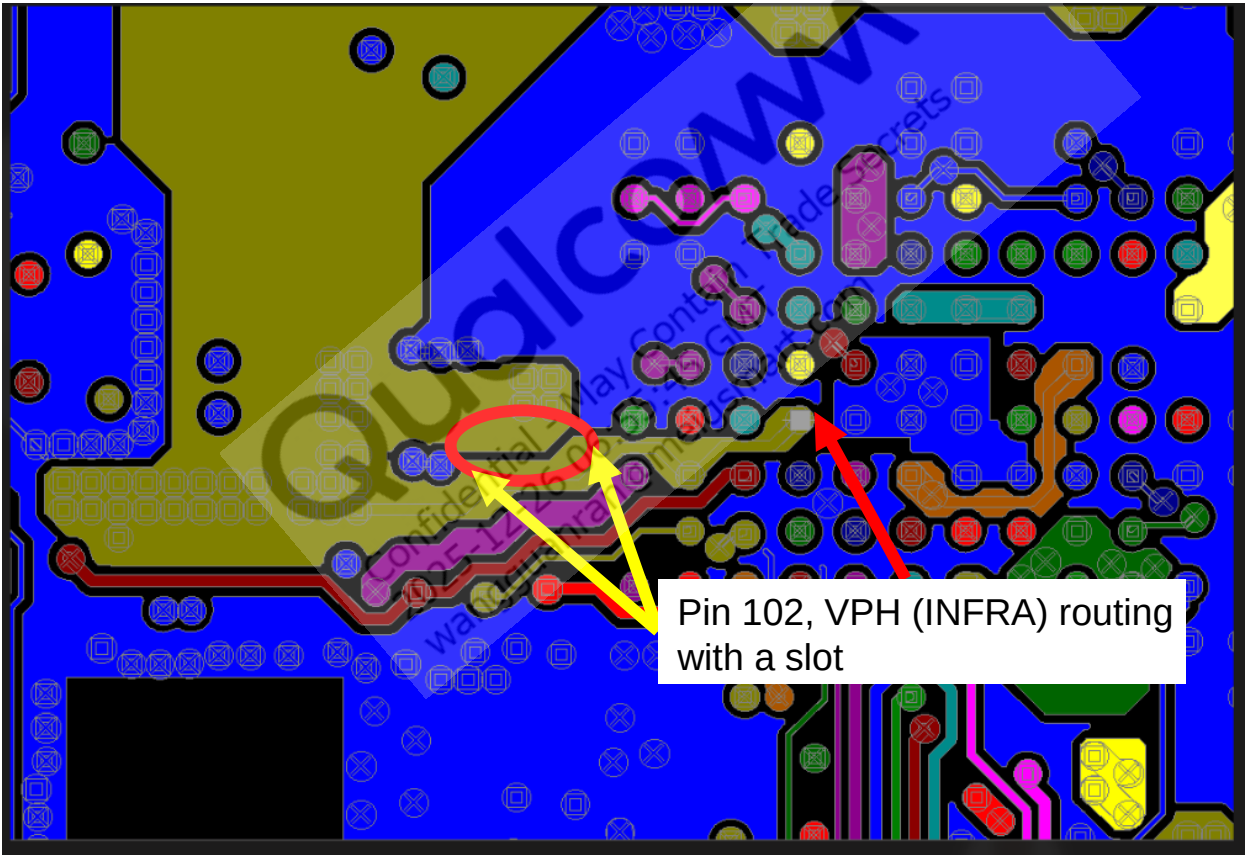


VPH μ Vias to VPH plane on L7

μ Vias to PMIC GND placed at VPH cap pads

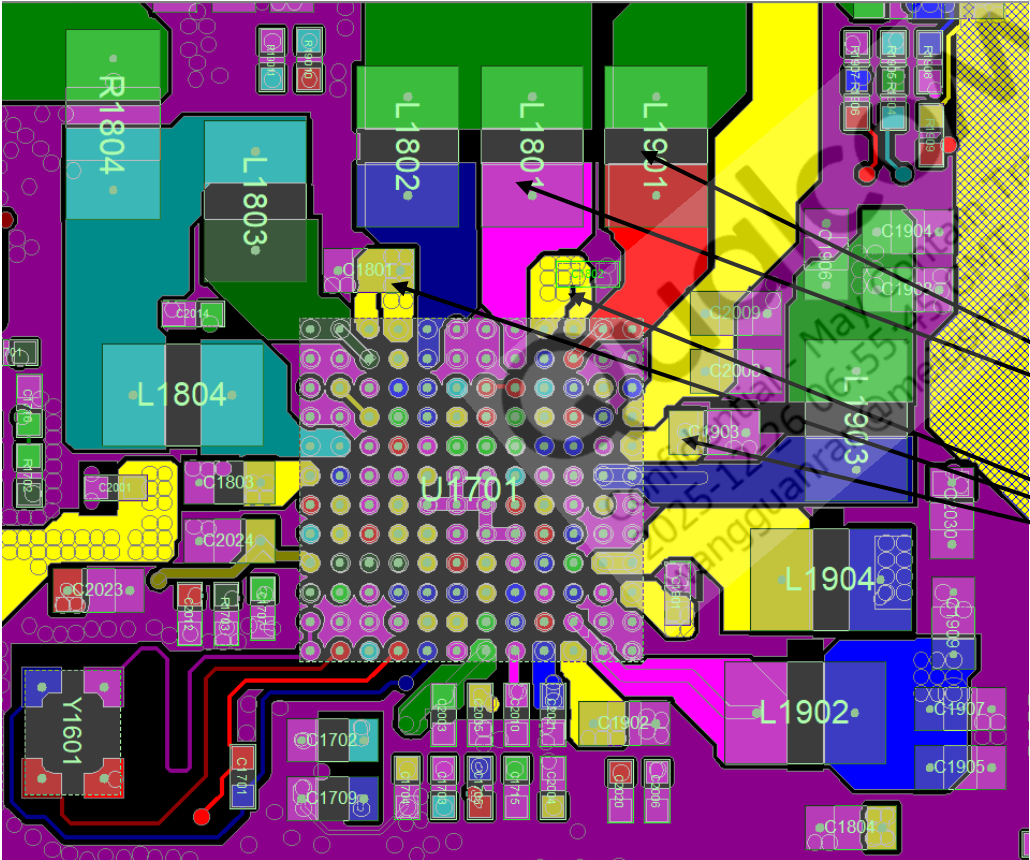
VPH Plane (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VPH (CMN)	The Zxy specifications against all SMPS VSW and VPH. The cross impedance, Zxy, should meet impedance specifications. Zxy is calculated with all capacitors populated and batfet open.	–	3 ----- 100	5 ----- 180	mΩ ----- pH	This is the reference voltage to the entire PMIC. Any noise that couples on to either signal will show up at all regulator outputs.



Component Placement Priority

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	VREG_Sx/ VREG_Lx/ INFRA	Passive component placement priority from closest to furthest from PMIC: 1) Buck Cin 2) Inductors 3) nLDO input/output caps 4) SMPS Cout for local sense 5) Capacitors for PMIC infrastructure 6) SMPS Cout for remote sense	—	—	—	—	Impedance requirements may not be met. Unnecessary noise propagation from aggressors such as VSW and VDD_Sx



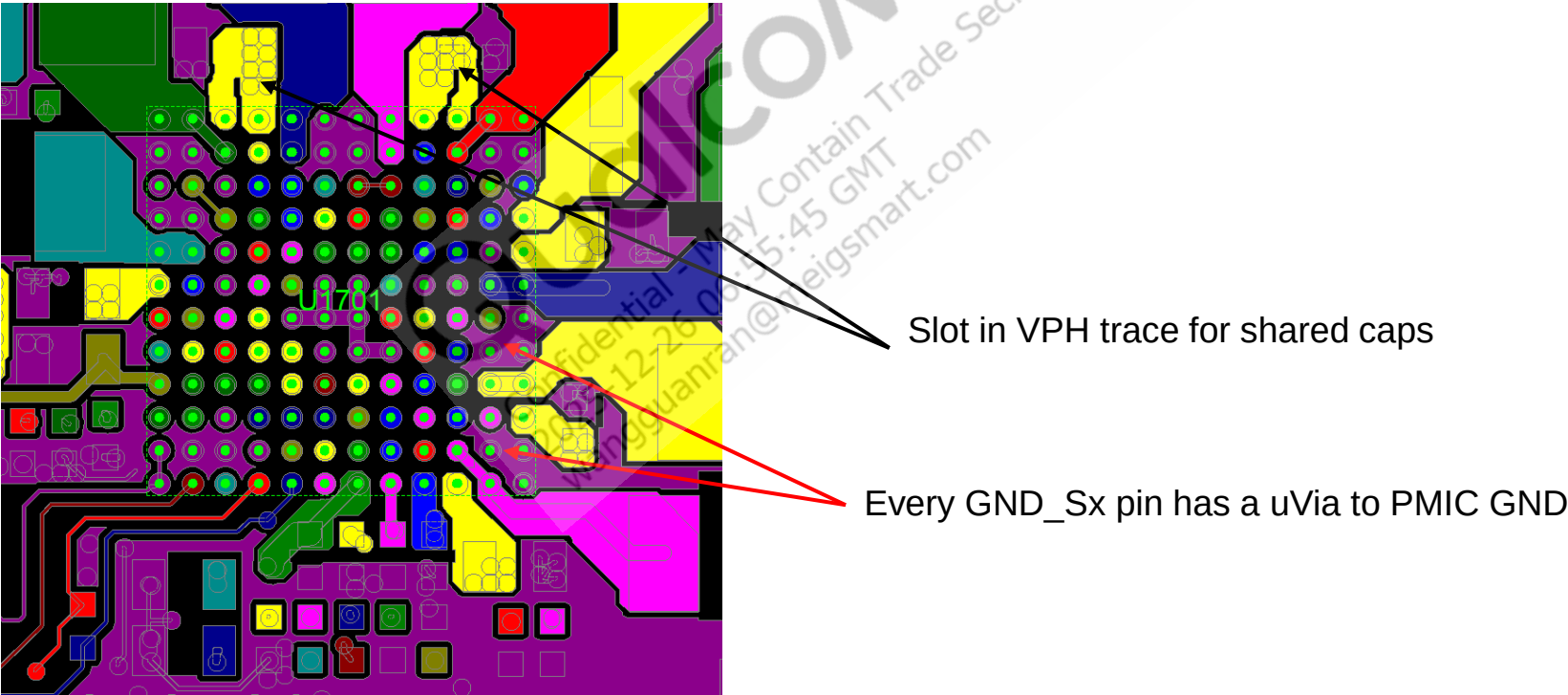
VCOIN

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Low	VCOIN	VCOIN trace must meet the DCR requirement.	–	100	200	mΩ	Too higher resistance will reduce the effectiveness of VCOIN when UVLO is hit. SMPL time window reduces (sudden momentary power loss)

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Buck Input

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VDD_Sx	Option 1: If VDD_Sx pins see a cap directly, route a direct trace to capacitor and have two μ Via to VPH plane located at capacitor pads. Option 2: If two buck inputs see one input capacitor, create a slot in the VPH trace to ensure cross impedance is reduced.	–	–	–	–	Insufficient μ Vias to VPH plane will increase Zxx.
High	GND_Sx	Each GND_Sx pin must have μ Via to layer 2 PMIC GND plane.	1	1	1	μ Via/pin	Noise may couple to other traces if these vias are located off-chip.

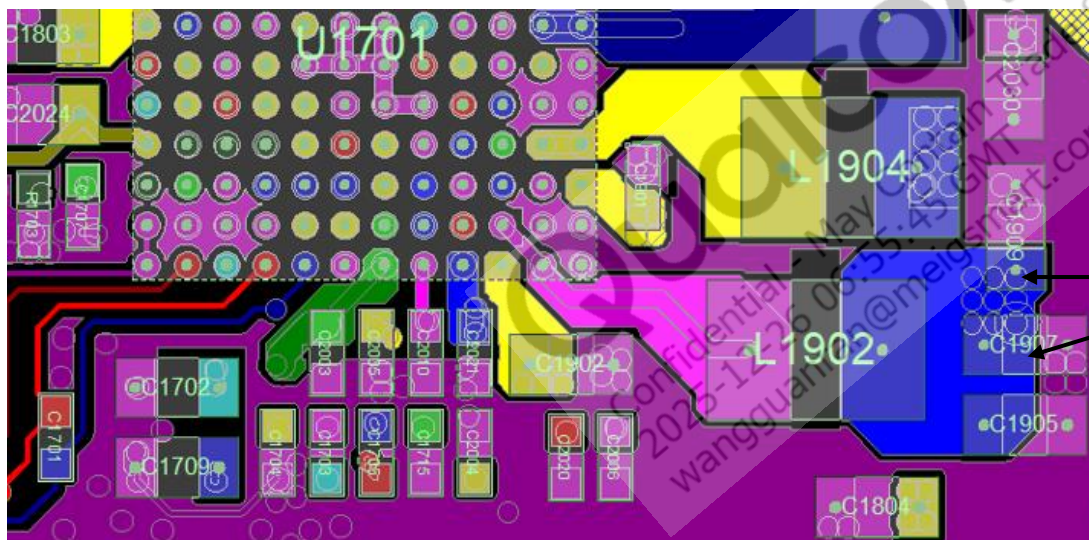


Buck Input (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VDD_Sx/GND_Sx	The self impedance of each buck input, Zxx, should meet the impedance specifications. Loop impedance from VDD_Sx pins to GND_Sx pins with VPH capacitors modelled.	See SM6225 Chipset Power Delivery Network Specification document (80-26986-1P) for target values			mΩ ----- pH	1. Voltage spikes resulting in damage to, or reduced lifetime of PMIC. 2. EMI due to radiation caused by high transient currents in the VIN_SX/PGND traces.
High	VDD_Sx/GND_Sx	To minimize the influence of buck over another, the cross impedance, Zxy, should meet the impedance specifications. Zxy is calculated with all capacitors populated and battfet open.				mΩ ----- pH	Reduced buck efficiency and performance due to mistriggering of buck comparators.

Buck Output

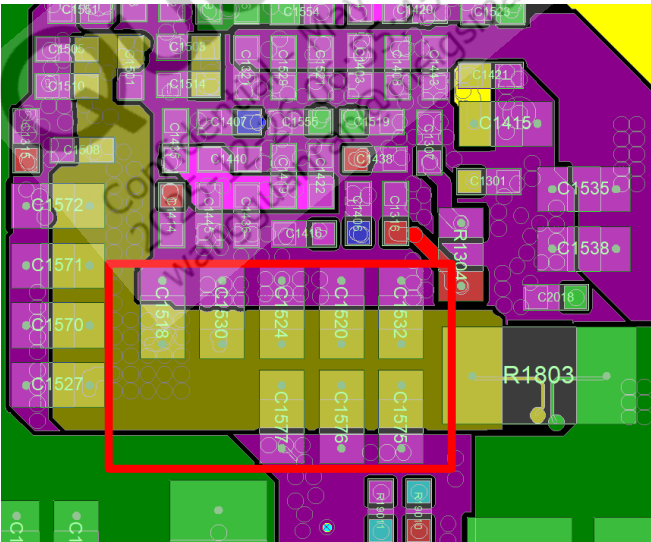
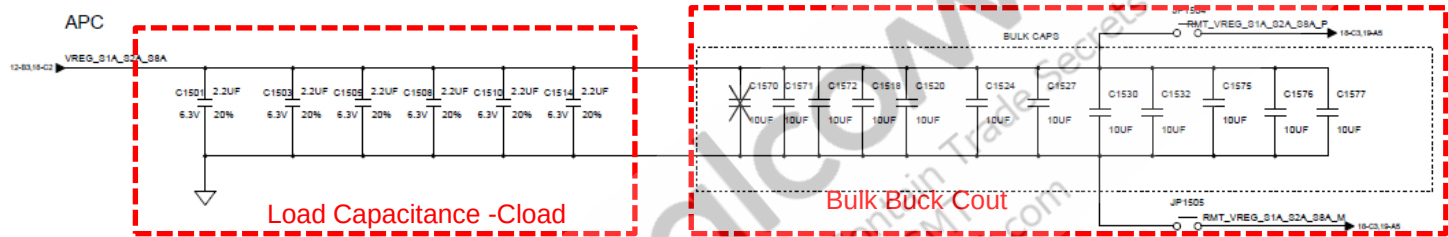
Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	VREG_Sx Output	Resistance from VSW pins to FB point with inductor shorted should meet the target for each of the buck individually. For multiphase bucks short all VSW pin to equipotential.	0	12	20	mΩ/phase	Reduced buck efficiency due to high PCB IR loss.
Medium	Buck Cout	All power routes must be star-routed from buck Cout. Use sufficient μVias at Cout pads to handle I _{max} . Power route for LDO input may be tapped before buck Cout in some cases. Contact QTI if there is a need for this.	2	3	5	μVia/pad	Taps before Cout will have higher noise and worse regulation.



- Sufficient μ Vias at Cout

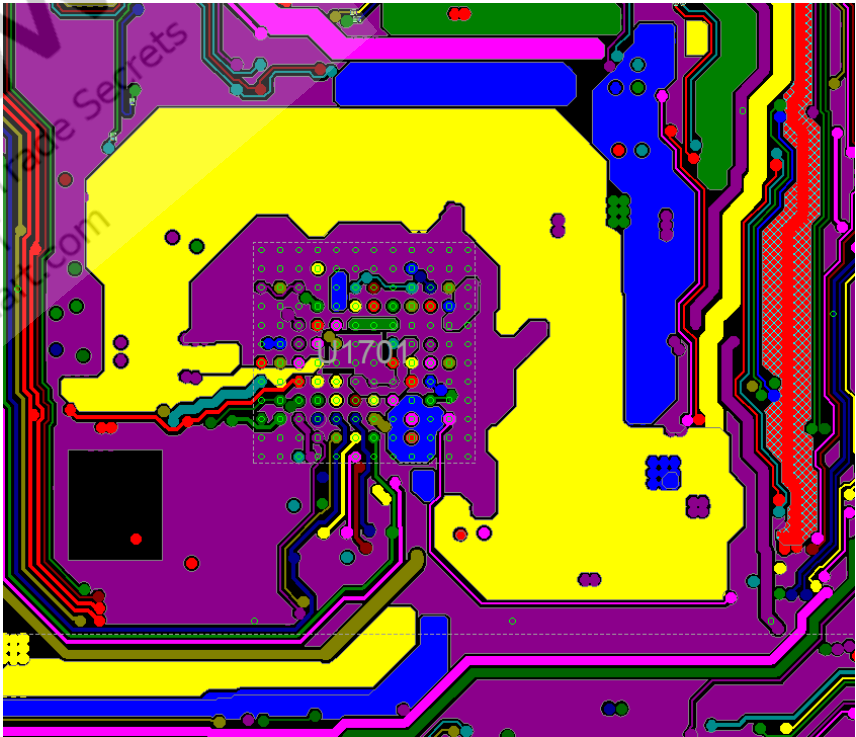
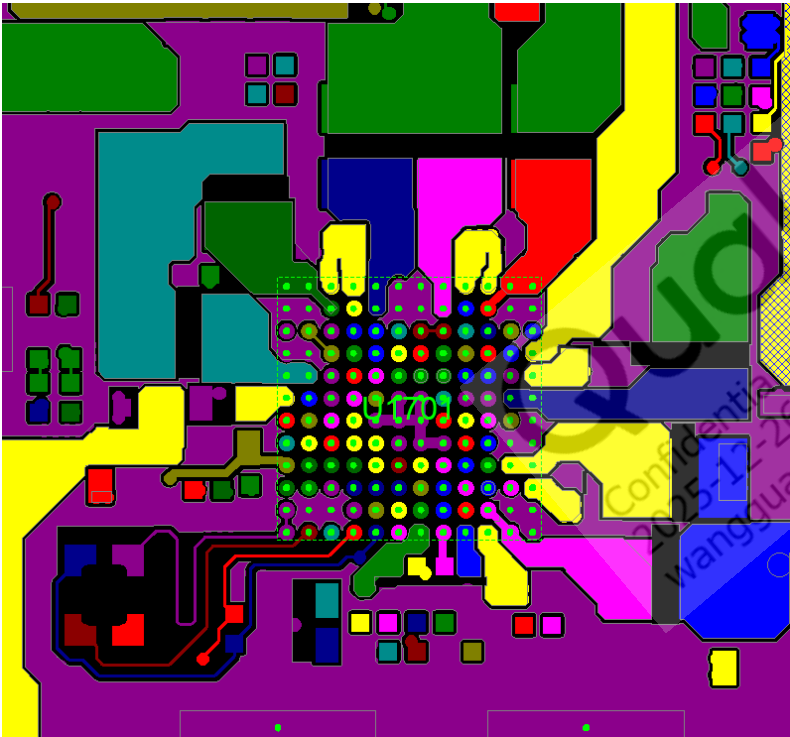
Buck Output (cont.)

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	Buck Cout	Impedance between any bulk buck Cout capacitors	0	0.05 ----- 100	5 ----- 500	mΩ ----- pH	Buck Cout should not be too far apart from each other, otherwise this may lead to buck instability and reduced efficiency.
High	FTS specific Cout + Cload	Total load capacitance distributed across all the loads must not exceed requirement (n = number of phases).	0	4 × n	10 × n	μF	Buck instability
High	FTS specific Cout + Cload	Ensure the impedance to any load capacitor from bulk buck Cout meets the requirement. Where the ratio of total cout per phase and total Cload. (x = Cout per ph/Cload total)	0	0.1x ----- 0.25x	0.75x ----- 1x	mΩ ----- nH	Buck instability



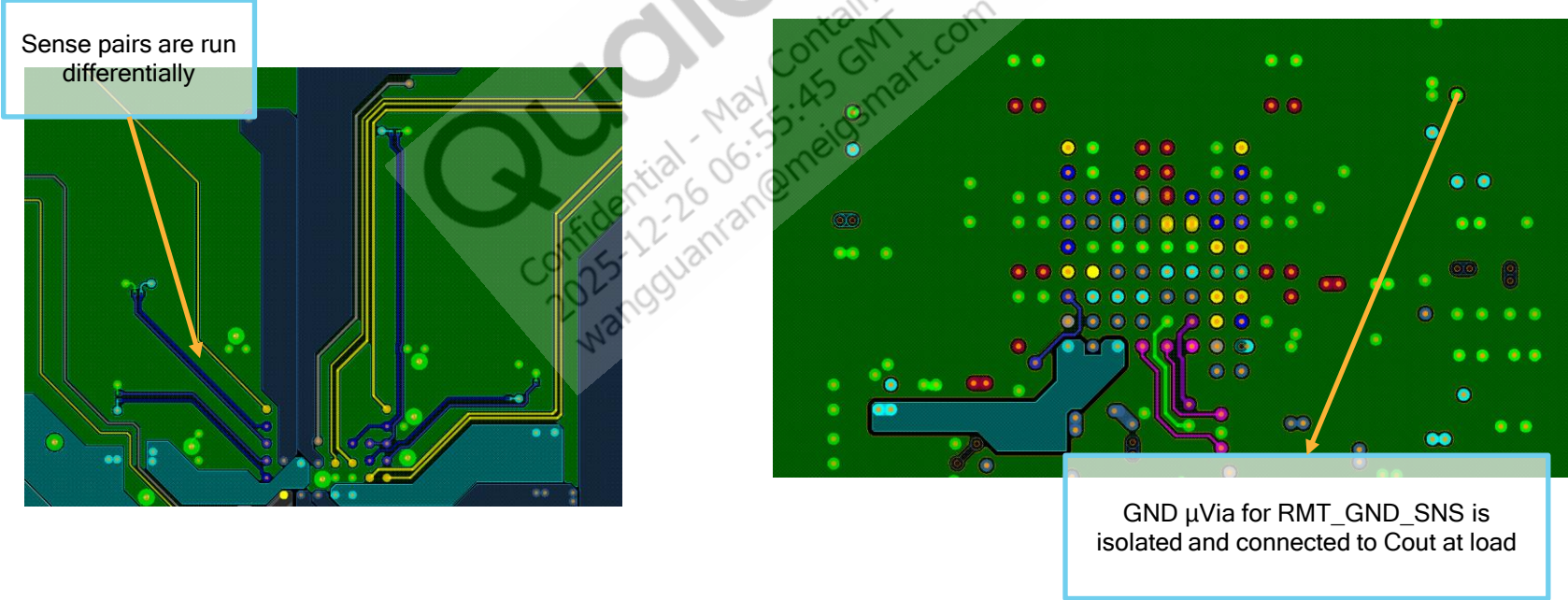
Buck VSW_Sx/VDD_Sx

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VSW_Sx/ VDD_Sx	Very noisy traces. They are strong aggressors. Any non-GND capacitive coupling to these traces should be kept within requirement.	0	0.5	1	fF	Sensitive signals in proximity will be impacted.



Buck Feedback

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_Sx_SNS and RMT_GND_SNS	FTS sense pairs must be routed differentially. Capacitive coupling between each other should meet specification. For multiphase domains, short sense pins at PMIC and only one sense pair should be broken out.	50	100	10000	fF	Reduced buck efficiency due to mistriggering of buck comparators.
Medium	VREG_Sx_SNS and RMT_GND_SNS	Tap SNS pair at or downstream of Cout with a dedicated, noncurrent carrying via. Resistance between SNS tap point and Cout (lumped) must meet the DCR requirements.	0	<1	3	mΩ	Taps before Cout will have higher noise and worse regulation and elevated output ripple.
Medium	HFS with remote FB	Impedance between inductor and remote FB point need to meet the impedance requirements.	0	1 ----- 0.1	33 ----- 5	% of DCR of inductor (mΩ) ----- % of inductance of inductor (pH)	Buck instability and reduced efficiency. Every 10 mΩ degrades efficiency by 0.6%.



Buck Feedback-HFS Remote Sensing

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	HFS with remote FB	Impedance between inductor and remote FB point need to meet the impedance requirements.	0	1 ----- 0.1	33 ----- 5	% of DCR of inductor (mΩ) ----- % of inductance of inductor (pH)	Buck instability and reduced efficiency. Every 10 mΩ degrades efficiency by 0.6%.

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LDO Input and Output

Priority	Net	Routing requirement	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_Lx	All LDO PDN requirements for input and output routes must meet the requirements listed in the respective chipset specific -1P document.	0	45–95	100	% of spec	Degraded LDO performance and functionality, for example, instability, degraded PSRR, and poor voltage regulation.

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VREG_XO

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_XO and GND_XO	Loop impedance from VREG_XO pin to GND_XO pin with VREG_XO cap shorted must meet the requirement.	–	20 ----- 0.5	50 ----- 3	mΩ ----- nH	Potential phase noise, phase, and frequency shift impact that will degrade the RF performance.
High	VREG_XO	Capacitive coupling from aggressors to VREG_XO must be within the specification.	–	–	1	fF	Noise from other grounds will couple into XO module, which will increase the associated CLKs' jitter and phase noise.

VREG_RF

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_RF and GND_RF	Loop impedance from VREG_RF pin to GND_RF pin with VREG_RF capacitor shorted must meet the requirement	–	20 ----- 0.5	50 ----- 3	mΩ ----- nH	Potential phase noise, phase, and frequency shift impact that will degrade the RF performance.
High	VREG_RF	Capacitive coupling from aggressors to VREG_RF must be within the specification.	–	–	1	fF	Noise from other grounds will couple into XO module, which will increase the associated CLKs' jitter and phase noise.

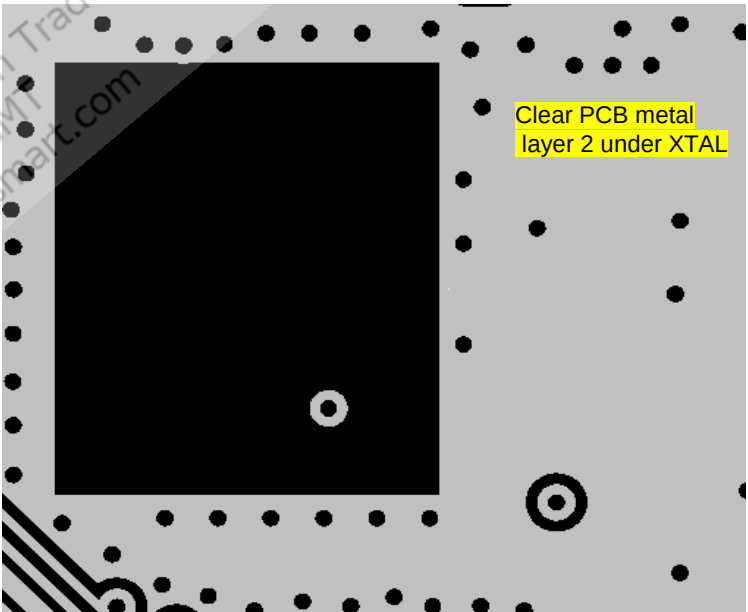
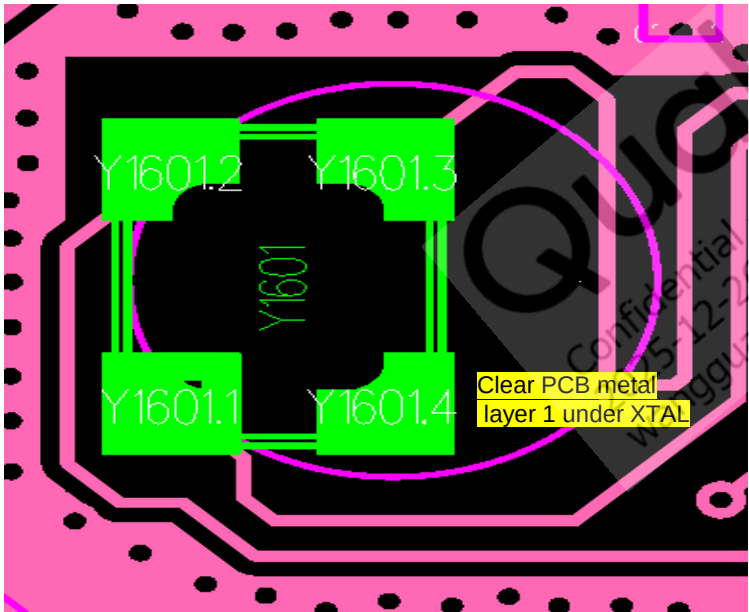
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VREG_BB and VIO_OUT

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	VREG_BB	Loop impedance from VREG_BB pin to GND_RF pin with VREG_BB capacitor shorted must meet the requirement.	–	20 ----- 0.5	50 ----- 3	mΩ ----- nH	Potential phase noise, phase, and frequency shift impact that will degrade the RF performance.
High	VREG_BB	Capacitive coupling from aggressors to VREG_BB must be within the specification.	–	–	5	fF	Noise from other grounds will couple into XO module, which will increase the associated CLKs' jitter and phase noise.
High	VIO_OUT	Loop impedance from VIO_OUT pin to its output capacitor must meet the requirement.	–	40 ----- 2	100 ----- 10	mΩ ----- nH	

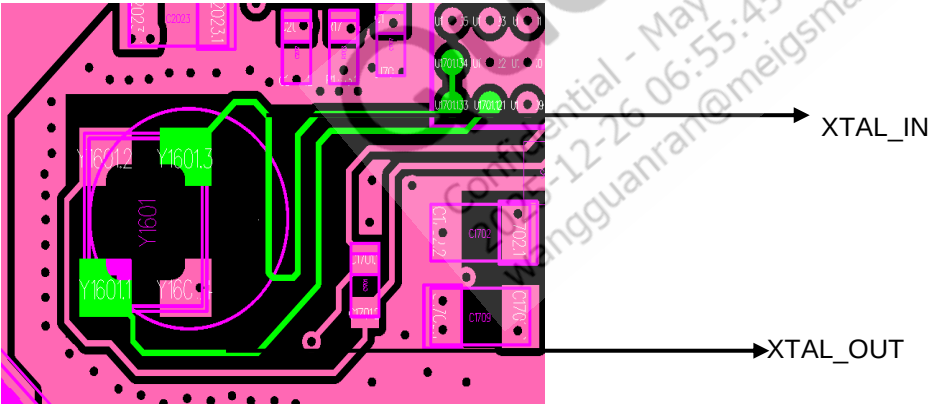
XTAL

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
High	XTAL	Clear PCB metal layers below the XTAL to reduce thermal transfer rate to XTAL.	90 ----- 1	100 ----- 1	100 ----- 2	% of XTAL perimeter ----- Number of layers	Sudden temperature changes of XTAL cause temporary frequency shifts leading to various RF performance problems such as dropped calls. See <i>Enhanced Guidelines to Implement 19.2 MHz and 38.4 MHz Crystals for Small PCB/High Thermal Layouts Application Note</i> (80-VP447-10) for more information.
High	XTAL	Place XTAL away from heat sources. Use the following guide: (SDM = 7–10 W, QPA = 2–4 W, camera flash = 4 W)	1	—	—	mm/W edge to edge	GPS TTFF (Time To First Fix) will be degraded during thermal transients. The GPS XO thermal compensation algorithm samples XO_THM at 10 Hz and can not tolerate greater than a 0.7°C/s ramp. Thermal simulations for a few industrial designs show that 1mm/W is a good target.



XTAL_IN/XTAL_OUT

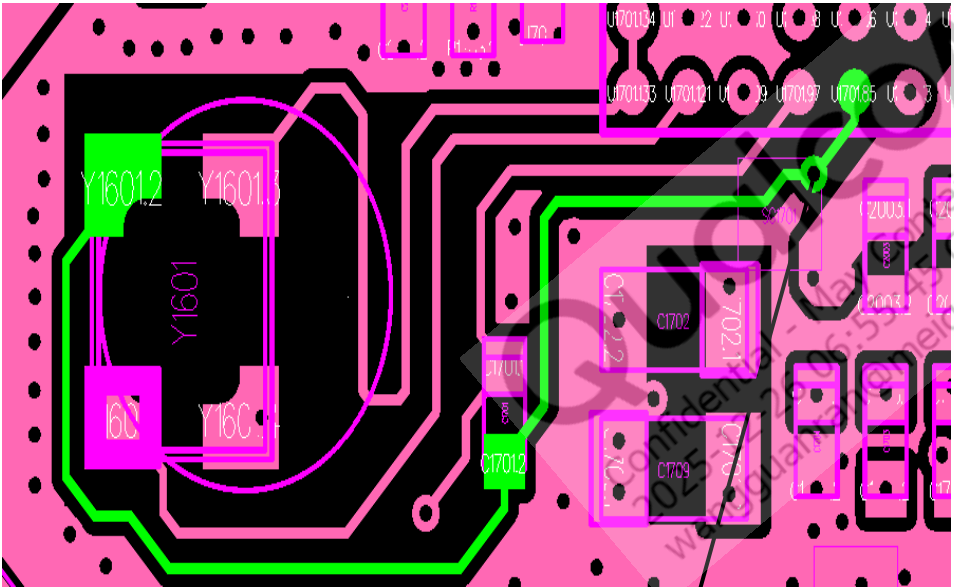
Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
High	XTAL_IN/ XTAL_OUT	Capacitive coupling from aggressors to XTAL_IN/XTAL_OUT must be within the specification.	–	–	0.1	fF	Results in instantaneous XO frequency spikes which WILL cause RF performance issues. Non-compliance has caused GPS parity errors, LTE call drops, RF de-sense and many other issues in the past. Results in degraded phase noise and jitter performance which can cause various RF performance issues, including degraded peak throughput. Phase noise specification violation have been seen on designs with coupling of 0.3 fF from RFCLK to VSW.
High	XTAL_IN/ XTAL_OUT	Cumulative capacitive coupling from all traces to XTAL_IN/OUT.	0	1.5	7	pF	Too much on board parasitic capacitance can cause the XO to run too slow and cannot be calibrated to the correct center frequency resulting in various RF performance problems such as dropped calls.



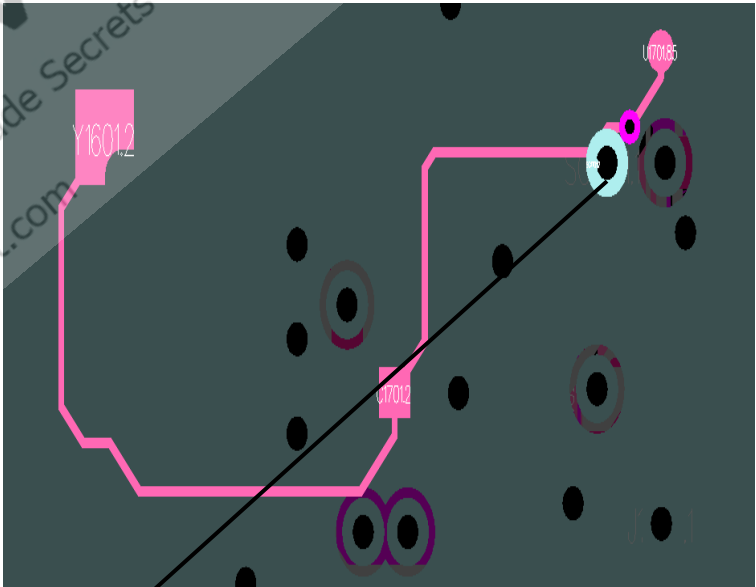
No aggressor in close proximity to avoid capacitive coupling

GND_XOADC

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
High	GND_XOADC	Route a minimum width trace to that PMIC. Add GND via at that PMIC.	0	1	2	mm	Grounding only at PMIC pin allows XTAL to be thermally isolated from GND plane. Grounding at XTAL pad can result in frequency shift which can cause various RF performance problems such as dropped calls.



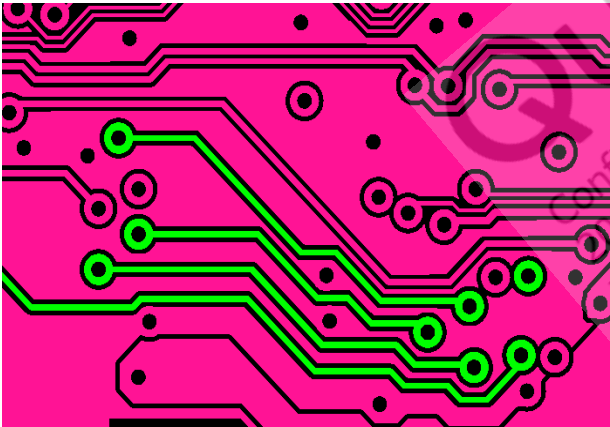
Via is placed near the PMIC pin



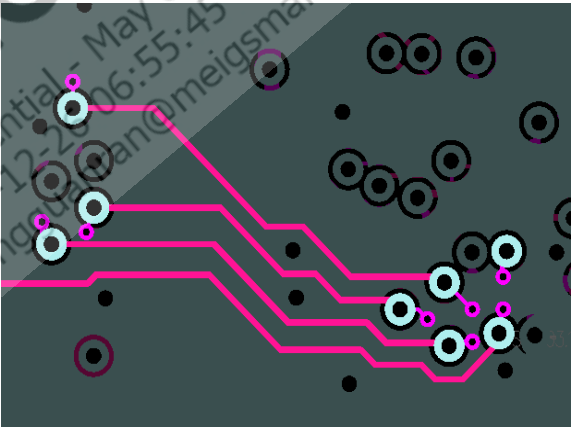
Connected to main ground plane in layer 4

RFCLKx and LNBBCLKx

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
High	RFCLKx/ LNBBCLKx	All CLK traces should be routed as strip line with GND. Capacitive coupling with regard to GND must meet the requirements.	3	6	10	pF	Phase noise degradation and power impact.
High	RFCLKx	RFCLKx routing capacitance, from smallest to largest, should be based on the following priority list (if present): 1. NFC 2. WCN If the required order is violated by the specified margin on the right, then place a capacitor near to CLK receiver of the highest priority to compensate for the difference in parasitic capacitance.	0	1	2	pF	Duty cycle calibration failure which will result in spurs and noise in RF transceiver and hence throughput degradation.
High	RFCLKx, LNBBCLKx	For each high frequency clock output, extract the PWB layout and run signal integrity simulation using QTI provided IBIS model to ensure SI compliance.	—	—	—	—	Poor signal integrity will impact clock phase noise and chipset functionality.



GND fill in-between the clock traces to avoid capacitive coupling between two clock signals



- Clock traces routed on layer 5, where layer 4 is GND plane and layer 6 has ground below these signals (strip-line routing)
- It will help to avoid noise coupling from switching nodes or other clock signals
- During RFCLKx and LNBBCLKx breakout if GND filling is not possible, provide three times trace width between the clocks.

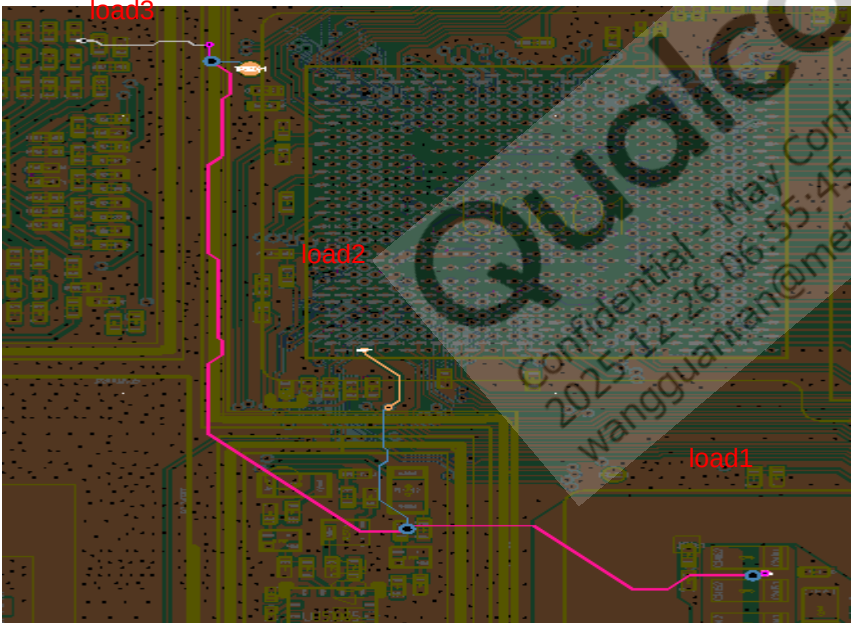
RFCLKx and LNBBCLKx (cont.)

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
High	RFCLKx/ LNBBCLKx	Capacitive coupling from aggressors to RFCLKx/LNBBCLKx must be within specification.	–	–	0.1	fF	Results in instantaneous XO frequency spikes which WILL cause RF performance issues. Non-compliance has caused GPS parity errors, LTE call drops, RF de-sense and many other issues in the past. Results in degraded phase noise and jitter performance which can cause various RF performance issues, including degraded peak throughput. Phase noise specification violation have been seen on designs with coupling of 0.3 fF from RFCLK to VSW.

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2025-12-26 06:55:45 GMT
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SLEEP_CLK

Priority	Pin	Description	Target			Units	Outcome of not meeting requirement
			Min	Typ	Max		
Medium	SLEEP_CLK	SLEEP_CLK can be routed in any one of the following three ways: 1) Star route 2) Daisy chain 3) Combination (star route + daisy chain) The longest path from CLK driver to CLK receiver needs to meet specification. (see section SLEEP_CLK Routing Impedance Guideline)	0	30	150	mm	Extensive SI simulations needed if specifications are violated. Sleep clock jitter and/or signal integrity might be impacted, which can lead to a missing triggering of the downstream clock receiver.

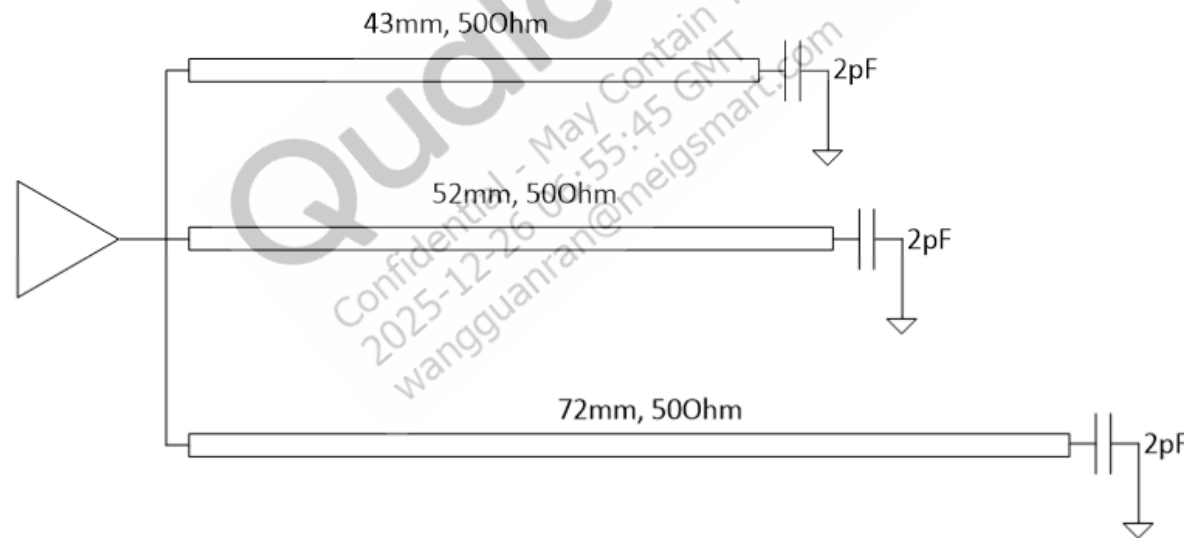


- Here, combination (star route + daisy chain) routing scheme is followed, see [SLEEP_CLK Routing Impedance Guideline](#) for more details on this
- Routed mainly on layer 5 to avoid noise coupling from aggressor

Sleep_CLK Layout Rule

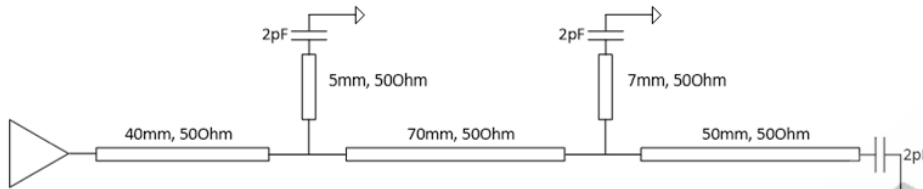
- Up to three clock receivers are allowed to be connected to the PM6225 sleep clock buffer using the guidelines provided below.
- The PCB traces to the clock receiver buffers should be 50 Ω , but can be routed in a daisy chain, star connect, or a combo scheme of routing as long as the routing length in each section is within the constraints set shown below:

1. Star route: In star route option, all clock traces split off from the driver right away. Length of the splits are preferred to be perfectly balanced (trace capacitor + load capacitor of all the traces should be equal). If the length cannot be balanced, mismatch (in absolute length) like what is shown below is tolerated.

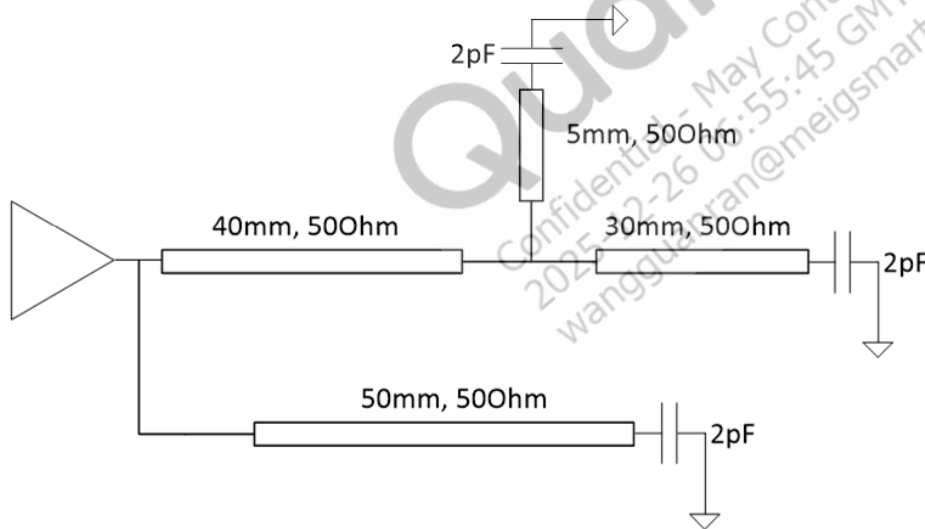


Sleep_CLK Layout Rule (cont.)

2. Daisy chain: In daisy chain scheme, a clock trace is routed to different clock receivers in a sequential manner. Recommendation for this scheme is listed below. Make sure each section is smaller or equal to the block diagram shown below.



3. Combination scheme: Sometimes, a combination of the two options above must be used. Make sure each section is smaller or equal to the block diagram shown



Questions?

For additional information or to submit technical questions, go to: <https://www.qualcomm.com/support>